

High frequency spike inference with particle Gibbs sampling

Reviewed Preprint

v2 • July 28, 2025

Revised by authors

Reviewed Preprint

v1 • March 27, 2024

Giovanni Diana , B Semihcan Sermet, Gerard J Broussard, Samuel S.-H Wang, David A DiGregorio 

Institut Pasteur, University of Paris, CNRS UMR 3571, Synapse and Circuit Dynamics Laboratory, Paris, France • Princeton Neuroscience Institute, Princeton, United States • Department of Physiology and Biophysics, University of Colorado School of Medicine, Aurora, United States

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

In their study, Diana et al. introduce a novel method for spike inference from calcium imaging data using a Monte Carlo-based approach, emphasizing the quantification of uncertainties in spike time estimates through a Bayesian framework. This method employs particle Gibbs sampling for estimating model parameter probabilities, offering accuracy comparable to existing methods with the added benefit of directly assessing uncertainties. The presentation of the underlying methods and its characterization is **convincing** and it presents a **valuable** advancement for neuroscientists interested in new approaches for parameter estimation from calcium imaging data.

<https://doi.org/10.7554/eLife.94723.2.sa3>

Abstract

Calcium-sensitive fluorescence indicators allow us to monitor the spiking activity of large neuronal populations in animal models. However, despite the plethora of algorithms developed over the past decades, accurate spike time inference methods for spike rates greater than 20 Hz are lacking. More importantly, little attention has been devoted to the quantification of statistical uncertainties in spike time estimation, which is essential for assigning confidence levels to inferred spike patterns. To address these challenges, we introduce (1) a statistical model that accounts for bursting neuronal activity and baseline fluorescence modulation and (2) apply a Monte Carlo strategy (particle Gibbs with ancestor sampling) to estimate the joint posterior distribution of spike times and model parameters. Our method is competitive with state-of-the-art supervised and unsupervised algorithms by analyzing the CASCADE benchmark datasets. The analysis of fluorescence transients recorded using an ultrafast genetically encoded calcium indicator, GCaMP8f, demonstrates our method's ability to resolve inter-spike intervals as short as five milliseconds. Overall, our study describes a Bayesian inference method to detect neuronal spiking patterns and their uncertainty. The use of particle Gibbs samplers allows for unbiased estimates of spike times and all model parameters, and it provides a flexible statistical framework to test more specific models of calcium indicators.

1 Introduction

Fluorescence indicators of calcium activity allow us to monitor the dynamics of neuronal populations both in vivo and in vitro. In the last decade there has been a proliferation of new methods to identify single spikes from fluorescence time series using template matching (Dyer et al., 2013; Grewe et al., 2010; Kerr et al., 2005; Quan et al., 2010), linear deconvolution (Holekamp et al., 2008; Tubiana et al., 2020; Wei et al., 2020; Yaksi and Friedrich, 2006), finite rate of innovation (Oñativia and Dragotti, 2015; Oñativia et al., 2013), independent component analysis (Mukamel et al., 2009), non model-based signal processing (Sebastian et al., 2019), supervised learning (Hoang et al., 2020; Rupprecht et al., 2021; Sasaki et al., 2008; Sebastian et al., 2021; Theis et al., 2016), constrained non-negative matrix factorization (Pnevmatikakis et al., 2014, 2016; Zhou et al., 2018), active set methods (Friedrich and Paninski, 2016; Friedrich et al., 2017), convex and non-convex optimization methods (Jewell and Witten, 2018; Jewell et al., 2020; Malik et al., 2011; Ranganathan and Koester, 2010; Stern et al., 2020), and interior point method (Vogelstein et al., 2010).

Model-based approaches aim to describe the mechanistic link between spikes and fluorescence. This is typically achieved by introducing a set of time-dependent “state” variables such as calcium level and baseline modulation, and their temporal dynamics. The temporal evolution of the state variables usually depends on additional parameters that remain constant over time and define the dynamics. Depending on the model, state variables might not be fully observable, in which case we need to infer them from the data. In statistics, this class of models is referred to as state-space models, and they are used in time series analysis to describe the probabilistic dependence between data and unobserved variables. Previous works have used tractable state-space models to derive maximum-a-posteriori estimates of spike times (Deneux et al., 2016; Fletcher and Rangan, 2014; Kazemipour et al., 2017; Tsunoda et al., 2010b).

The vast majority of spike inference methods provide single estimates of spike times by minimizing a cost function defined by the underlying model and constraints. This optimization approach does not provide information about the statistical uncertainty associated with single-point estimates. To address this issue, previous studies proposed Bayesian inference methods (Greenberg et al., 2018; Huys and Paninski, 2009; Im et al., 2019; Mishchenko et al., 2011; Mishchenko and Paninski, 2011; Pnevmatikakis et al., 2013; Rahmati et al., 2016; Shibue and Komaki, 2020; Speiser et al., 2017; Theis et al., 2013; Tsunoda et al., 2010a; Vogelstein et al., 2009) which give access to the full probability distribution of the unknowns given the data.

However, the state-space models used in these approaches do not take into account the possibility of burst firing and slow changes in the fluorescence baseline, which is known to be an important issue for the analysis of in-vivo recordings. Moreover, current Bayesian methods do not treat time-independent model parameters (e.g. rate constants) and dynamic variables equally. Instead, they require additional optimization procedures to calibrate model parameters, typically relying on ad-hoc tuning or grid search. This separation can lead to biased inference and poorly calibrated uncertainty estimates, particularly when parameters such as calcium decay time or spike amplitude are inaccurately specified. In contrast, our approach jointly infers both spike times and model parameters within a unified Bayesian framework, enabling uncertainty-aware estimation and avoiding separate, error-prone calibration steps.

Inference on non-linear and non-Gaussian state-space models is analytically intractable, requiring the application of Monte Carlo methods to obtain unbiased approximations of the posterior distributions. Because the number of unknowns in these models is of the order of the number of time steps in the fluorescence time-series, the analysis of long time series requires efficient strategies to sample from high-dimensional spaces. A major breakthrough in analyzing state-space

models has been the introduction of sequential Monte Carlo methods (Chopin and Papaspiliopoulos, 2020). These algorithms can sample efficiently from the latent space by approximating sequentially the target posterior distribution by combining importance sampling and resampling techniques. In particular, the particle Gibbs algorithm can be used to obtain unbiased estimates of the joint distribution of time-independent model parameters and dynamical variables, but it has never been applied in the context of spike inference.

In this work we employ the particle Gibbs (PG) sampler on a bursting autoregressive (BAR) model of time series calcium-dependent fluorescence to provide not only point estimates of spike times but also quantify the statistical uncertainty of each estimate. This is important for downstream analyses such as comparing activity across neurons or conditions. Our generative model accounts for periods of high firing rates between periods of baseline (lower) firing rates. By quantifying the performance of our method (PGBAR) on the CASCADE benchmark dataset (Rupprecht et al., 2021) we showed that our approach is competitive with existing techniques. Finally, we tested PGBAR on in-vitro recordings of cerebellar granule cells using the ultrafast GCaMP8f calcium indicator, showing that our method permits the detection of spikes reliably even for high firing rates ($\sim 200\text{Hz}$).

2 Results

2.1 The model

To infer spike times and their uncertainty from noisy fluorescence traces, we first build a probabilistic generative model that captures the main dynamics underlying the fluorescence signal. We model the normalized fluorescence $F_{1:T}$ as the sum of a calcium-dependent fluorescence level c_t (hereinafter referred to as calcium level for brevity), a time-varying baseline b_t and Gaussian noise η_t

$$F_t = c_t + b_t + \eta_t, \quad t = 1, \dots, T \quad (1)$$

where the fluorescence noise η_t is normally distributed with zero mean and variance σ^2 . During in vivo neural recordings, animals can display different behaviour and are usually exposed to different sensory stimulations. Neural activity can potentially reflect this non-stationary activity and alter their firing rates during the recording time. To account for varying firing rate, in our model we make the simplifying assumption of a bimodal firing rate, described as a hidden two-state process to separate periods of high and low firing rate. Therefore we introduce firing states $q_t = 0, 1$, associated respectively with low and high firing rates, r_0 and r_1 . We allow for stochastic transitions between these two states with rates $w_{0 \rightarrow 1}$ and $w_{1 \rightarrow 0}$. The probability of switching from q to q^i within a sampling period Δ is given by the transition matrix

$$W = \begin{bmatrix} 1 - w_{0 \rightarrow 1}\Delta & w_{0 \rightarrow 1}\Delta \\ w_{1 \rightarrow 0}\Delta & 1 - w_{1 \rightarrow 0}\Delta \end{bmatrix} \quad (2)$$

The number of spikes at time t , s_t , is modeled by a Poisson distribution with rate r_1 when $q_t = 1$ otherwise with baseline firing rate r_0 when $q_t = 0$. The dynamics of the calcium level in response to a spike train is modeled as a second order autoregressive process

$$c_t = \begin{cases} c_0 + As_1 & t = 1 \\ \gamma_1 c_1 + As_2 & t = 2 \\ \gamma_2 c_{t-2} + \gamma_1 c_{t-1} + As_t & t > 2 \end{cases} \quad (3)$$

where c_0 is the initial calcium level and A controls the calcium increase upon single action potential. Note that in Eq. (3) the calcium level at time t depends on the previous calcium levels up to c_{t-2} . The dynamics of c_t in response to a single spike (kernel response) is characterized by a

finite rise time (time to peak response) and exponential decay (see Section Response kernel for a derivation).

We can recast our model as a first order Markov process, where the state at time t only depends on the state at time $t-1$. This can be done by introducing a calcium vector and a spike count vector (see S2.2.3 in Ref. Pnevmatikakis et al. (2016) [\[1\]](#))

$$C_t = \begin{bmatrix} c_t \\ c_{t-1} \end{bmatrix}, \quad S_t = \begin{bmatrix} s_t \\ 0 \end{bmatrix}, \quad (4)$$

where in particular the calcium vector at time t is constructed by combining the calcium levels at current and previous time. With this definition, the calcium vector C_t satisfies the first-order Markov dynamics

$$C_t = \begin{cases} [c_0 + As_1, 0] & t = 1 \\ M \cdot C_{t-1} + AS_t & t > 1 \end{cases}, \quad M = \begin{bmatrix} \gamma_1 & \gamma_2 \\ 1 & 0 \end{bmatrix}. \quad (5)$$

In this model the calcium dynamics is a deterministic process given the sequence of spikes $s_{1:T}$.

We reparameterize the autoregressive model in terms of phenomenological parameters: peak response ($A^{(max)}$), rise time (time to peak response τ_r) and decay time (τ_d) of unitary fluorescence response. Thanks to previous characterization of GCaMP probes (Chen et al., 2013 [\[2\]](#)), we can more easily design prior distributions for such phenomenological parameters rather than the autoregressive model parameters γ 's, A and $\gamma_{1,2}$. $A^{(max)}$, τ_r and τ_d , referred to as kernel parameters in the upcoming sections, can be derived from $\gamma_{1,2}$ and A as (see Section Reparameterization for a derivation)

$$A^{(max)} = A \cdot g_A, \quad g_A \equiv \left(\frac{g_+}{g_-} \right)^{\frac{g_+}{g_-}} \left(1 - \frac{g_+}{g_-} \right) (e^{g_+} - e^{g_-})^{-1} \quad (6)$$

$$\tau_r = \frac{\log \left(\frac{g_+}{g_-} \right)}{g_- - g_+}, \quad g_{\pm} = \log \left(\frac{\gamma_1 \pm \sqrt{\gamma_1^2 + 4\gamma_2}}{2} \right) \quad (7)$$

$$\tau_d = -\frac{1}{g_+}, \quad (8)$$

Finally, the fluorescence baseline B_t is described by a Gaussian random walk with normally distributed initial condition

$$\begin{cases} B_t \sim \mathcal{N}(0, 1) & t = 1 \\ B_t \sim \mathcal{N}(B_{t-1}, \sigma_B^2 \Delta) & t > 1 \end{cases} \quad (9)$$

where Δ is the sampling period of the time series.

In the language of state-space models, the latent state of our model is the combination of the bursting state q_t , the spike count s_t , the calcium vector C_t and the baseline b_t , whereas the fluorescence F_t , defined in Eq. (1) [\[1\]](#) is our observation. The time-independent parameters of our model are the firing rate constants $r_{0,1}$, the transition rates of the 2-state bursting process $W_{0 \rightarrow 1}$, $W_{1 \rightarrow 0}$, the kernel parameters of the calcium indicators (peak amplitude, rise and decay constants), the initial calcium level c_0 and the fluorescence noise σ . To simplify the notation we will denote the latent space as $X = \{q_t, s_t, C_t, b_t\}$ and the combination of time-independent parameters as θ .

2.2 State-space model formulation

The joint probability of the latent state trajectory $X_{1:T}$ and the fluorescence observations $F_{1:T}$ conditional to the time-independent parameters θ can be expressed as

$$P(X_{1:T}, F_{1:T}|\theta) = \mu^\theta(X_1) \cdot \prod_{t=2}^T f_t^\theta(X_t|X_{t-1})g_t^\theta(F_t|X_t) \quad (10)$$

where $f_t^\theta(X_t|X_{t-1})$ is the transition probability of the latent state, $g_t^\theta(F_t|X_t)$ is the probability of the observed fluorescence conditional to the latent state at time t and $\mu^\theta(X_1)$ is the probability distribution of the initial latent state. The latent state transition probability can be expressed in terms of calcium level, firing state and baseline transitions and the Poisson probability of spike counts, namely

$$f_t^\theta(X_t|X_{t-1}) = \overbrace{\delta^{(2)}(C_t - M \cdot C_{t-1} - AS_t)}^{\text{deterministic calcium}} \cdot \overbrace{W_{q_{t-1}q_t}}^{\text{firing state}} \cdot \overbrace{\frac{(r_{q_t}\Delta)^{s_t}}{s_t!} e^{-r_{q_t}\Delta}}^{\text{Poisson spikes}} \cdot \overbrace{(2\pi\Delta\sigma_b^2)^{-1/2} \exp\left(-\frac{1}{2\Delta\sigma_b^2}(b_t - b_{t-1})^2\right)}^{\text{baseline}}. \quad (11)$$

By assuming the fluorescence noise to be normally distributed, we have

$$g_t^\theta(F_t|X_t) = (2\pi\sigma^2)^{-1/2} \cdot \exp\left[-\frac{1}{2\sigma^2}(F_t - c_t - b_t)^2\right]. \quad (12)$$

To infer latent states and time-independent parameters from fluorescence observations, we need to compute the posterior probability

$$P(X_{1:T}, \theta|F_t) = \frac{P(\theta) \cdot P(X_{1:T}, F_{1:T}|\theta)}{P(F_{1:T})}, \quad (13)$$

where $P(\theta)$ denotes the prior probability on model parameters and $P(F_{1:T})$ is the normalization factor of the posterior distribution, also known as marginal likelihood. This distribution encodes all the information about the statistics of the latent state trajectory and the model parameters. We can use it to compute point estimates but also to quantify uncertainties. The posterior distribution for general state-space models is not analytically tractable. However, we can use Monte Carlo methods to generate random samples from the posterior distribution and use them to obtain unbiased approximations of any posterior average.

2.3 Sequential Monte Carlo

The model described in the previous section is analytically intractable, therefore we employ Monte Carlo methods to sample from the posterior distribution in Eq. (13) of spike times and model parameters, allowing us to make probabilistic inferences rather than relying on point estimates alone. There are two critical issues when applying these methods to state-space models. First, the high-dimensionality of the posterior distribution and, second, the joint inference of state variables and constant parameters. In the general setting of a state-space model we need to estimate state variables at each time point. Consequently, the support of the target posterior distribution is a high-dimensional space. The high dimensionality of state-space models is an issue when applying standard sampling methods such as Markov Chain Monte Carlo, as their performance rapidly decreases with increasing dimensionality. Sequential Monte Carlo (SMC) methods address this issue by providing efficient sampling strategies from the latent space. The typical approach is to construct a sequential approximation of the posterior distribution where observations are accounted for iteratively. SMCs solve the problem of estimating dynamical variables at fixed model parameters (filtering).

The second issue, related to the joint estimation of parameters and dynamic variables, was addressed by Andrieu et al. (2010) [with](#) the introduction of the particle Gibbs algorithm. This algorithm alternates the sampling of model parameters and state variables as in the Gibbs sampler, with the difference that state variables are sampled from an SMC-based transition kernel that leaves the filtering distribution $P(X_{1:T} | F_{1:T}, \theta)$ invariant. In this work, we employ a version of the particle Gibbs algorithm developed by Lindsten et al. (2014) [named](#) particle Gibbs with ancestor sampling (PGAS), with better mixing properties (see Algorithm 2 and the Methods section).

To carry out inference of time-independent and dynamical variables for our model, we employed Algorithm 1, which alternates the two steps mentioned above: (1) run the PGAS transition kernel to sample a new trajectory of the state variables, and (2) sample model parameters from their full conditional distributions when analytically available, otherwise from a Metropolis-Hastings kernel.

2.4 Validation and performance of PGBAR on simulated time-series fluorescence data

To test the performance of our inference method, we generated latent state variables and fluorescence time series from our model and compared the spikes inferred using our sampling algorithm against the ground truth simulations. In

Fig. 2A [we](#) show a fluorescence time series simulated from our model. The firing pattern displays periods of increased firing rate separated by quiet time windows. By using this trace as input to Algorithm 1, we can generate a latent state trajectory $X_{1:T} = \{q_t, s_t, C_t, b_t\}_{t=1}^T$ and a set of model parameters at each iteration. In **Fig. 2B** [we](#) show 1000 samples of spike counts obtained by fitting the normalized fluorescence in **Fig. 2A** [. The](#) average spike counts over the random samples at each time frame (**Fig. 2C** [\) can be interpreted as the instantaneous firing rate multiplied by the sampling period. To illustrate the accuracy of our method, we calculated the spike counts within 1s time intervals for each random sample, providing the posterior distribution of the number of spikes in each time bin. As shown in Fig. 2D](#) [, the ground-truth spike counts are well within the range of the posterior. Our method allows us to infer not only spike times but also the time windows of high and low firing states \(\$q_t = 0, 1\$ in the model\) of the neuron. In particular, the probability of a burst-firing state \(\$q = 1\$ \) can be obtained by averaging the firing state across the Monte Carlo samples. As shown in Fig. 2E](#) [, this probability is close to one during the ground-truth bursting periods and zero otherwise, with some degree of uncertainty at the onset and offset of the bursting period. Fig. 2F](#) [compares the ground-truth baseline and the sample average.](#)

Algorithm 1 Gibbs sampler

```

1: Set  $\theta^{(1)}$  and  $X_{1:T}^{(1)}$ 
2: for  $n > 1$  do
3:   draw  $X_{1:T}^{(n)} \sim \mathcal{K}_{\theta^{(n-1)}}^N(X^{(n-1)}, \cdot)$  (PGAS kernel)
4:   draw  $\theta^{(n)} \sim P(\theta | X_{1:T}^{(n)}, F_{1:T})$ 

```

One of the key advantages of our sampling algorithm is the joint estimation of latent states and time-independent model parameters, such as spike amplitude, decay time, noise level, and baseline variance. This stands in contrast to most existing spike inference algorithms, which rely on fixed or externally calibrated parameters. Such fixed-parameter methods are vulnerable to systematic errors when parameter values are uncertain or misestimated. By jointly sampling from the posterior of all variables and parameters, our method propagates uncertainty correctly and mitigates bias due to manual tuning or poor initialization.

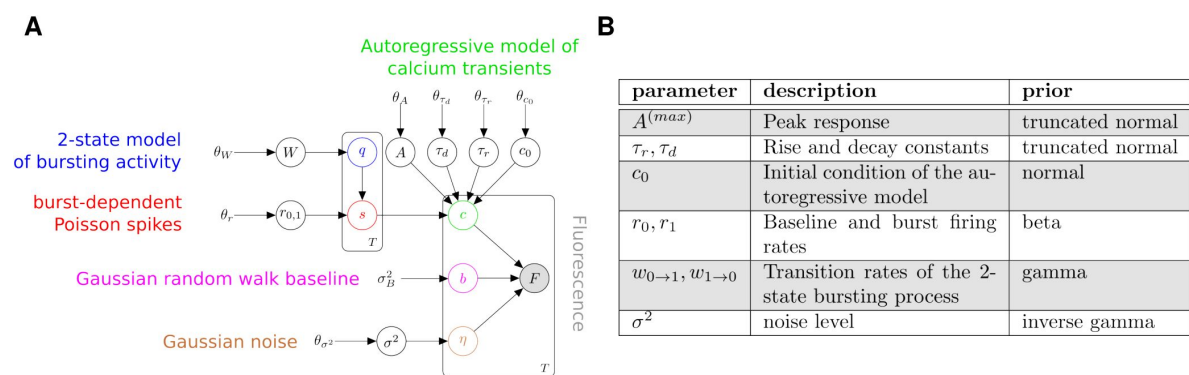


Figure 1.

Generative model of fluorescence time series

(A) Graphical representation of the generative model described in the main text. White circles denote unknown variables, grey circles denote measurements and bare variables are fixed prior hyperparameters. Plates denote groups of variables. **(B)** List of parameters and corresponding priors.

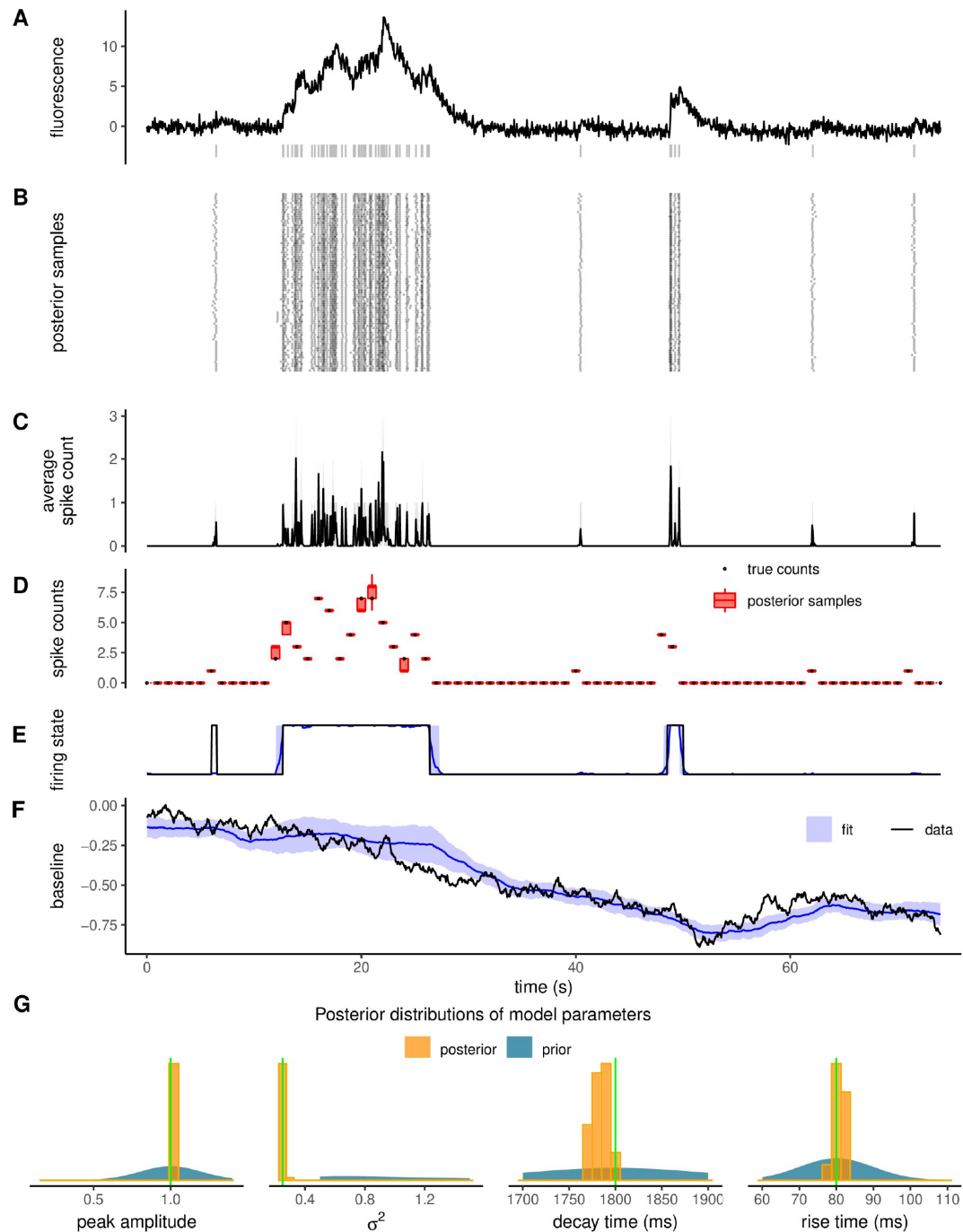


Figure 2.

Validation of the spike inference approach with simulated data

(A) Example trajectory simulated from the model (solid, black) with ground-truth spike times shown underneath (grey vertical lines). (B) Raster plot representing spike times for a thousand Monte Carlo samples. (C) Average spike counts over the Monte Carlo samples at each time frame. (D) Comparison between ground-truth counts over 1s bins (black dots, from the example trace in A) and the corresponding posterior distributions (red boxes). (E-F) Comparison of ground-truth firing state and baseline (solid, black) to estimated ones (blue). Shading indicates one standard deviation from posterior averages. (G) Posterior distributions of peak response upon a single spike, decay time, rise time, and noise level compared with true value (vertical lines in green).

Figure 2G [↗](#) illustrates the posterior distributions associated with the fluorescence probe's peak amplitude, noise level, decay, and rise time. The ground truth parameters used to simulate the testing time series are always close to the peak of the corresponding posterior distributions, showing that our model is identifiable.

To quantify the importance of having two firing states on the accuracy of the inference, we compared the performance of our method against a variant with only one global Poisson firing rate. We simulated fluorescence traces at different signal-to-noise (SNR) levels (1.1, 2 and 10), defined as the ratio between peak response $A^{(max)}$ and the fluorescence noise parameter σ , and the burst firing frequency parameter, r_1 (5Hz, 10Hz, 20Hz, 50Hz). Then, we used our algorithm and its non-bursting variant to infer spikes from the fluorescence trace. To quantify the inference performance, we calculated the correlation between ground truth and estimated spikes (downsampled at 7.5 Hz for consistency with other analyses in this work), the average absolute error and the bias (average error) per time point (see Methods). The top panel of **Figure 3A** [↗](#) shows two example traces with stimulation times generated by a 5 and 50Hz Poisson distribution of spike times. For the 5Hz firing trace, the bursting model did not improve the inference accuracy compared to the variant with a single firing rate (**Fig. 3A** [↗](#), middle and bottom). The two analyses produced comparable correlations, errors and biases (**Fig. 3C-E** [↗](#)). In contrast, the single firing rate model induced a systematic underestimate of the number of spikes for the 50 Hz trace, the original model captures the ground-truth spikes reliably.

The lower performance of the non-bursting version of the model is due to the bias induced by forcing a single firing rate across the time series. While for all conditions of noise and frequency, inference using the bursting model gives unbiased spike counts (**Fig. 3E** [↗](#)), in the case of the non-bursting model, the single Poisson firing rate leads necessarily to an underestimation of the spike count during bursting time windows and an overestimation during low activity windows. At increasing noise levels and firing frequency, the performance difference between bursting and non-bursting versions of our algorithm become more pronounced (**Fig. 3D** [↗](#)), with a clear advantage of our original bursting model in increasing correlation with ground truth and reducing error.

2.5 Validation of PGBAR on the CASCADE benchmark data and comparison to previous methods

To test our method on experimental data we analyzed neuronal recordings from the CASCADE benchmark dataset ([Rupprecht et al., 2021](#) [↗](#)), which allowed us to quantify the performance of our algorithm on different calcium indicators. Bayesian priors for all PGBAR parameters were adapted to each experiment according to the existing characterisation of the different calcium indicators used ([Chen et al., 2013](#) [↗](#)). We compared our method to CASCADE ([Rupprecht et al., 2021](#) [↗](#)), MLSpike ([Deneux et al., 2016](#) [↗](#)), Peeling ([Lütcke et al., 2013](#) [↗](#)), CaImAn ([Giovannucci et al., 2019](#) [↗](#)), Suite2p ([Pachitariu et al., 2017](#) [↗](#)) and JewellWitten ([Jewell et al., 2020](#) [↗](#)) by using their previously benchmarked performance on the same datasets obtained from extensive parameter optimization ([Rupprecht et al., 2021](#) [↗](#)) (**Fig. 4C** [↗](#)). As a metric for inference performance, we used the Pearson's correlation coefficient between ground-truth spikes and predicted spikes after filtering with a Gaussian kernel with 200 ms bandwidth. This metric allowed us to directly compare PGBAR with previous analyses [Rupprecht et al. \(2021\)](#) [↗](#). The correlation obtained with PGBAR averaged across cells and datasets is 0.75. As for the other methods, we observed a large variability of performance across recordings, however we did not find a particular condition where our method performed systematically better or worse across indicators (**Fig. 4A** [↗](#)) and standardized noise level (**Fig. 4B** [↗](#)), which is a noise index introduced by [Rupprecht et al. \(2021\)](#) [↗](#) to account for different sampling frequencies (ratio between the standard deviation of the normalized fluorescence and the square root of the sampling

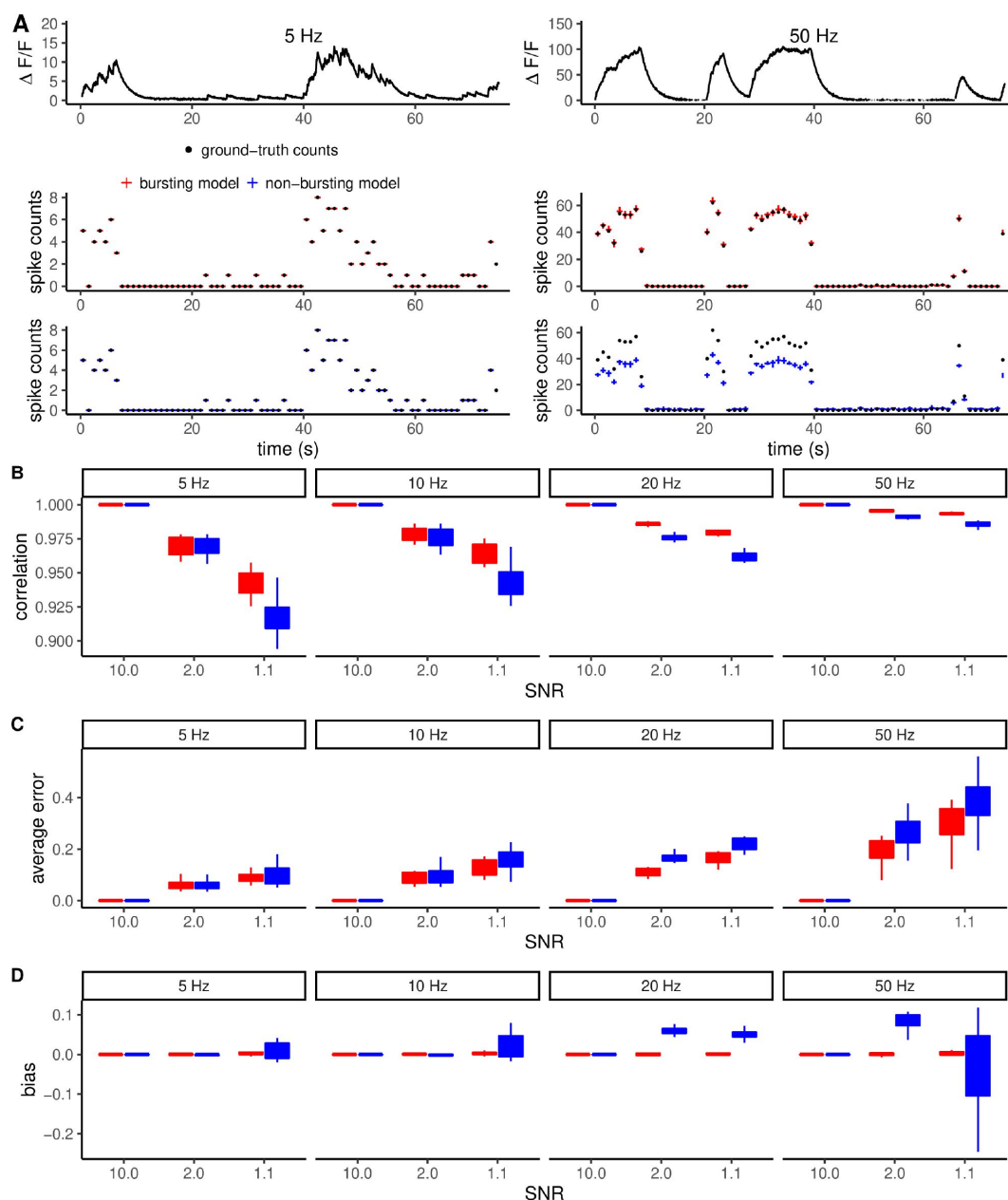


Figure 3.

Dependency of inference performance on noise and firing frequency and the bias of non-bursting models

(A) Example fluorescence time series simulated at 5Hz and 50Hz bursting frequencies (top). The analysis of these traces using the bursting and non-bursting variant of the model highlights the large bias generated by the non-bursting model at high frequency (bottom). **(B-D)** Quantification of correlation with true spikes, average error, and bias at different levels of SNR and frequency. At increasing firing frequency, the correlation with ground-truth spikes generally increases. This is an effect of calculating correlations at fixed temporal resolution. The average error was quantified as the sum of the absolute deviation from the true spike counts divided by the number of time steps.

frequency). In particular, we did not observe a statistically significant difference between mean correlation coefficient across standardized noise levels in the range 0.5-3.5 (**Fig. 4B**), although lower noise levels seem to be associated to a larger range of correlation.

The performance range of PGBAR across the CASCADE database was comparable to previous methodologies. Among model-based approaches, our method slightly underperformed MLSpikes (PGBAR median is within 6% from MLSpikes, Mann-Whitney test p-value 0.0043), likely due to its description of the non-linear properties of the calcium indicator. The top performance is achieved by the supervised CASCADE method, however, it is a supervised method (i.e. needs training on ground truth data) and does not provide posterior distributions of spike times and model parameters.

The previous analysis provides an overview of the general performance of PGBAR in comparison with previous methods. To illustrate the advantages of PGBAR in estimating statistical uncertainties, we now take a closer look at a representative recording in the CASCADE dataset (**Fig. 5A**, CASCADE dataset 9 (Chen et al., 2013)), a GCaMP6f fluorescence time series from a pyramidal neuron in the mouse visual cortex. The comparison between ground-truth spikes and those inferred using PGBAR in **Fig. 5A** shows differences outside the 1st-3rd interquartile range in 30% of the 1s time intervals. This statistical discrepancy between the posterior distributions and the ground truth can be attributed to the limitations of the autoregressive model to reproduce the biophysical properties of GCaMP6. Despite such discrepancy, the estimated bursting pattern shown in **Fig. 5A** (lower panel) captures faithfully the overall periods of increased neuronal activity. The posterior distributions of model parameters are shown in **Fig. 5B**. Some distributions (burst firing rate and the rise time) shift significantly from their corresponding priors. The mismatch between prior and posterior also highlights the shortcomings of the model. In this particular case, the posterior distribution of the burst firing rate appears shifted to larger values. This result could be attributed to the use of Gaussian noise in the model. Indeed, non Gaussian fluorescence fluctuations might be erroneously interpreted by the sampler as true spikes, pushing the burst firing rate to larger values. This example illustrates that prior distributions can be used to partially compensate this bias by penalizing unrealistic parameter configurations.

2.6 Validation and performance of PGBAR on simulations of short-interval stimuli

We tested the accuracy of PGBAR in resolving pairs of stimuli with inter-spike intervals (ISI) below 10ms by performing model simulations of different signal-to-noise levels. **Figure 6A** shows two simulated fluorescence traces with two spikes 10ms apart at low (1.4) and high (3.4) SNR levels. By running our algorithm on these simulated data PGBAR was accurately detecting 2 spikes with over 95% confidence across all SNR values. Next, we focus on the temporal accuracy of the ISI estimation. Increasing SNR or sampling frequency has the effect of narrowing the ISI posterior distribution around the ground-truth value (**Fig. 6B**). When examining multiple stimulus intervals, SNRs, and sampling frequencies, we show that high sampling frequencies enable more reliable extraction of spike intervals. The posterior probability of the ISI falls within 3 ms of the ground-truth ISI at sampling frequency of 1 kHz and within 6 ms for 3 kHz (averaging over 30 trials, see **Fig. 6C**). When this probability is above 0.5, the ground-truth value is equal to the mode of the posterior ISI distribution (using 3 ms bin size), therefore we can use this probability as a metric to quantify the temporal accuracy of our inference. For ground-truth values above 3 ms, the contours of constant probability (iso-probability) show that increasing the temporal separation between stimuli has no effect on the inference accuracy. On the contrary, increasing the sampling frequency shifts the iso-probability contours towards lower SNR, indicating that for high sampling rates, lower SNRs are required to reach the same accuracy. **Figure 6D** shows the posterior ISI distributions obtained from ten independent simulations with ground-truth ISI of 5 ms at different SNR levels. At low SNR the posterior distributions have larger variance while at higher SNR levels

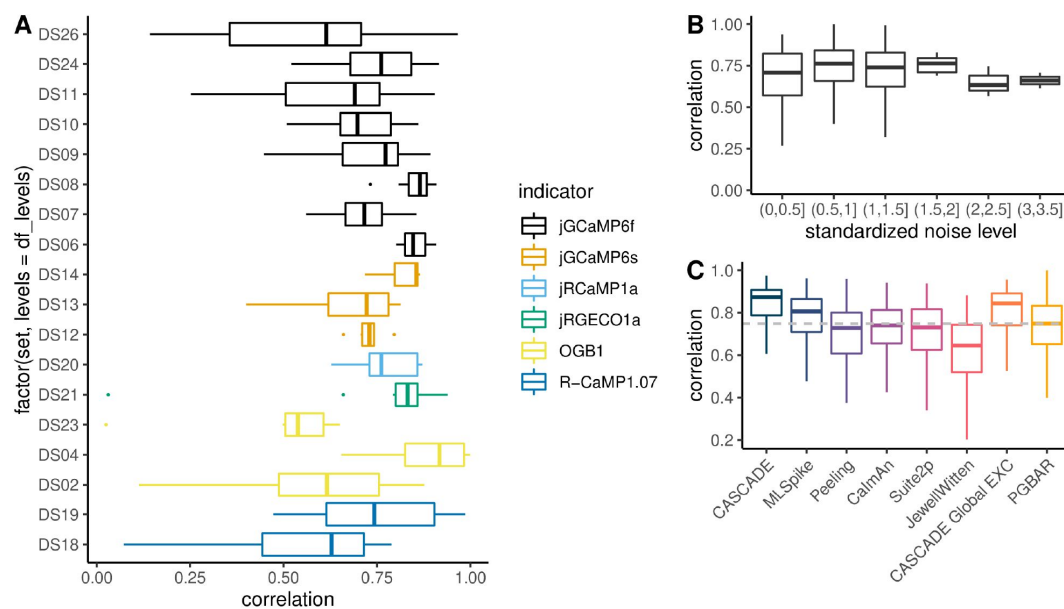


Figure 4.

Comparison of PGBAR with existing methods using analysis of CASCADE benchmark data

(A) Correlation between estimated and ground-truth firing rates filtered with a 200ms bandwidth Gaussian kernel (CASCADE dataset). The color code represents the different calcium indicators employed in each dataset. **(B)** Correlation with ground-truth spikes as a function of the standardized noise level (Rupprecht et al., 2021). **(C)** Comparison with existing methods. Correlation averaged across datasets and neurons.

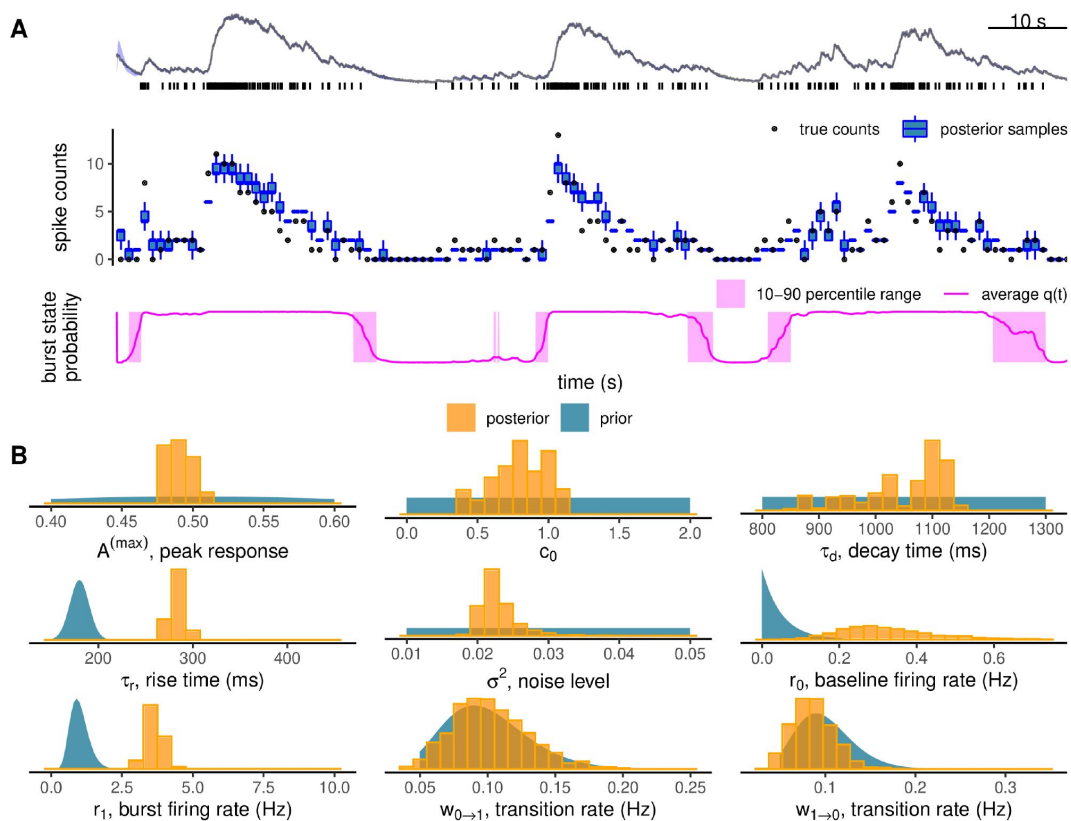


Figure 5.

Analysis of GCaMP6f recordings from the CASCADE dataset

(A) example $\Delta F/F$ from the CASCADE dataset (#DS09, GCaMP6f, mouse visual cortex) with ground-truth spikes shown underneath fluorescence (top), comparison of spike counts within 1s time intervals (middle) and burst probability (bottom). Shading denotes uncertainty within one standard deviation. **(B)** Comparison between posterior distributions of the model parameters (histograms) and priors (continuous densities): maximal calcium response to single spikes ($A_{\max}$), initial calcium level (c_0), decay and rise time, noise level (σ^2), bursting (r_1) and baseline (r_0) firing rates, transition rates between firing states ($w_{0 \rightarrow 1}$, $w_{1 \rightarrow 0}$).

they shrink around the ground-truth value. This analysis illustrates the expected variability of spike inference when analyzing multiple trials of the same neuron. Based on this analysis we expect PGBAR to provide accurate estimates of inter-spike intervals down to 5 ms.

2.7 PGBAR spike inference from fluorescence transients recorded using the fast calcium indicator GCaMP8f

We tested our approach on the fast calcium indicator GCaMP8f by performing high-speed (≈ 2.8 kHz) two-photon linescan calcium imaging of cerebellar granule cells *in vitro*. GCaMP8f was expressed in the Crus I region of the cerebellum using adeno-associated virus (AAV) injection (Fig. 7A). Compared to GCaMP6f, GCaMP8f exhibits a rise time that is nearly an order of magnitude faster, which we expected to translate into substantially improved temporal accuracy in spike time detection. Fluorescence signals were recorded from both granule cell somata and synaptic boutons along the parallel fibers, while ground-truth spikes were evoked via extracellular stimulation of granule cell axons in the molecular layer (Fig. 7B).

For each recorded soma and bouton we applied two types of stimuli. Single time point stimulation and a fixed stimulation pattern generated from a 20 Hz Poisson process with 29 stimulation time points. First, we used the single stimulation trials to design prior distributions of amplitudes, rise and decay constants (Fig. 7C). Next, we used PGBAR to analyze independently each Poisson stimulation trial in Figure 7E. By generating thousands of posterior samples of spike time patterns, we obtained the spike probability for all time frames and trials (Fig. 7F). The Pearson's pairwise correlation between spike probabilities (Gaussian filtered with 20 ms bandwidth) across trials is always larger than 0.95, which demonstrates that PGBAR provides robust predictions across trials and it can reliably detect single-trial action potential-evoked GCaMP8f fluorescence transients.

To illustrate the temporal accuracy of PGBAR we focused on a part of the stimulus train with a short 5 ms interval between two spikes (Fig. 7G). Despite the relatively low SNR ($A^{(max)}/\sigma \approx 2.4$), 100% of the posterior samples contained two spikes in the considered time interval. The ground-truth ISI is well within the posterior distribution of each trial, with posterior modes symmetrically distributed around it (Fig. 7H). To better understand the variability of the posterior modes, we simulated 10 fluorescence traces with two spikes separated by 5 ms and sampling frequency matching our recordings. By analysing these simulations with PGBAR we obtained a distribution of ISI posterior modes that is statistically compatible with the real data (Fig. 7I, Mann-Whitney test p-value = 0.97), which provides further evidence that our results are statistically unbiased and that PGBAR can be used to infer spike times for intervals down to 5 ms.

The distribution of the posterior modes of the total number of spikes across experiments and trials is centered around the ground-truth value for both somas and boutons with a relative error of 15% (Fig. 8A). To quantify the precision of spike time detection, we used a single trial definition of temporal accuracy as the offset between ground-truth spikes and nearest detections averaged across stimulation time points. This temporal accuracy is 0.43 ms (± 0.10 ms, SEM) and 1.39 ms (± 0.11 ms, SEM) in boutons and somas respectively, highlighting a significant delay of 0.95 ms (± 0.15 ms SEM, Mann-Whitney test p-value = 4.8×10^{-7}) from the average time of detection in boutons (Fig. 8B). This result is compatible with the proximity of synaptic boutons to the electrical stimulation along the parallel fiber. We analyzed both signals from somata and synaptic boutons because *in vivo* two-photon imaging can be performed from both parts of the cell. Here we showed that our method performs reliably on both, demonstrating its robustness across recording sites.

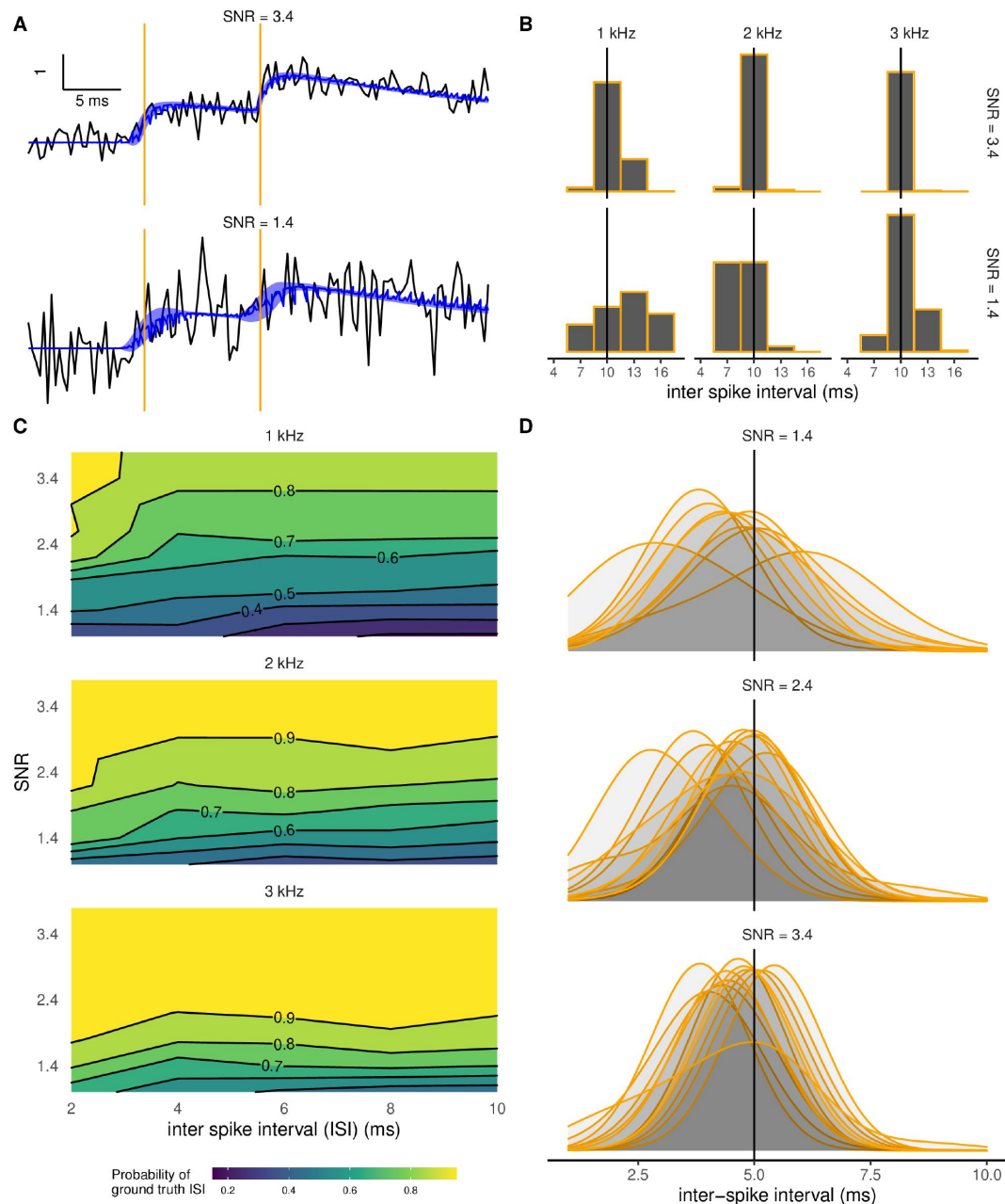


Figure 6.

Sensitivity of spike detection to sampling frequency and SNR level

(A) Examples of simulated fluorescence traces with two spikes separated by 10 ms (vertical lines) at low SNR (1.4, bottom) and high SNR (3.4, top). Shaded bands display denoised fits (calcium fluorescence plus baseline) within one standard deviation. (B) Posterior distributions of the inter-spike interval (ISI). Increasing SNR (from bottom to top) and sampling frequency (left to right) has the effect of reducing the ISI posterior variance, bringing the maximum-a-posteriori estimate (MAP) closer to the ground-truth. (C) Posterior probability of the ISI to be within an interval of 3 ms centered around the ground truth ISI as a function of SNR and ground-truth ISI with sampling frequency of 1, 2 and 3 kHz. (D) Trial-to-trial variability of ISI posterior distributions. We analyzed 12 simulated fluorescence traces with sampling frequency of 3 kHz and two spikes separated by 5 ms. Density plots have been smoothed with 1 ms bandwidth. For all simulations we used $\tau_r = 3.7$ ms and $\tau_d = 40$ ms.

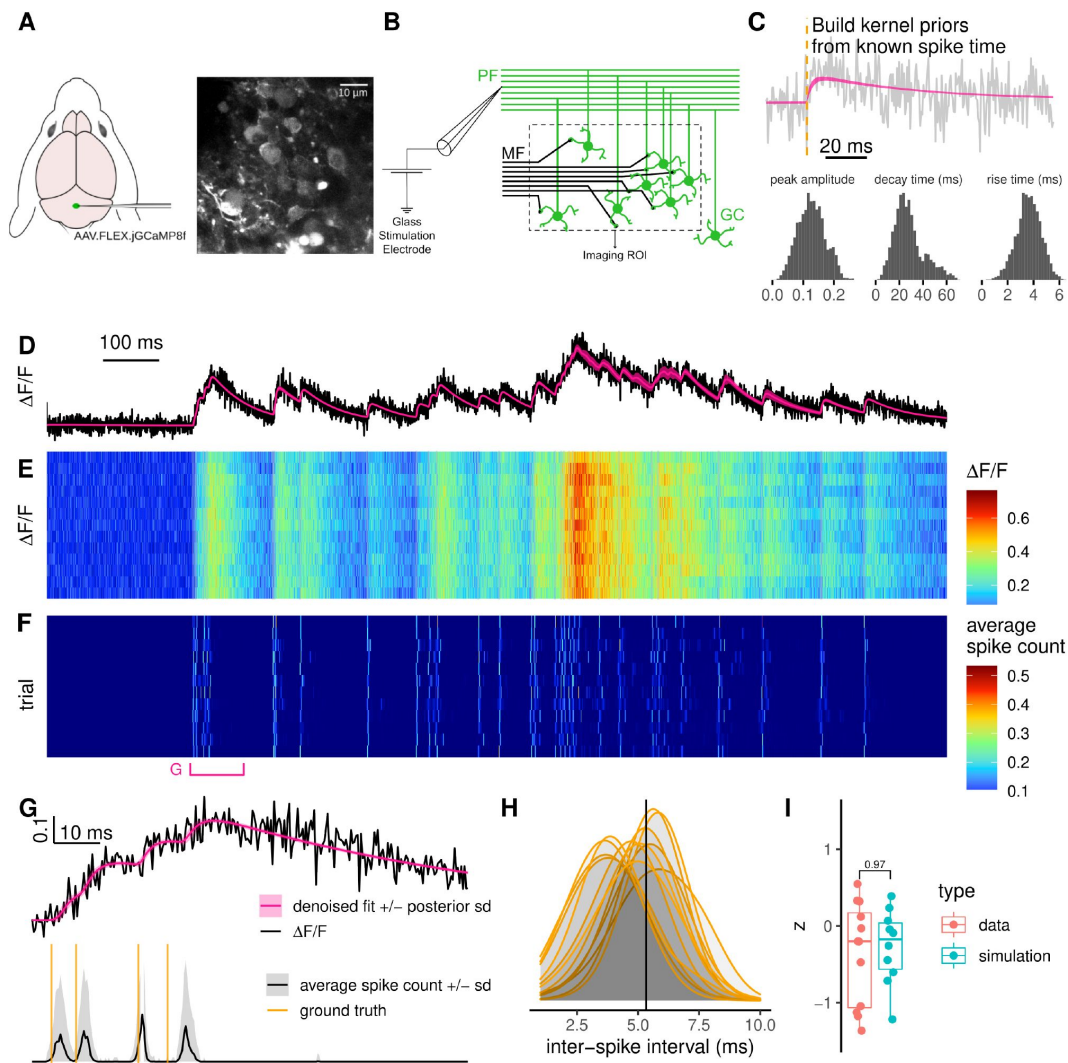


Figure 7.

High-speed 2-photon linescan calcium imaging

(A) GCaMP8f virus injection in the cerebellar vermis. (B) Induction of action potentials to cerebellar granule cells by direct stimulation of the parallel fibers. (C) Single pulse stimulation to extract kinetic model parameters of GCaMP8f indicator. (D-E) High-speed (3 kHz) 2-photon linescan calcium imaging of granule cell somata. (D) Representative fluorescence time series from a single trial and (E) heatmap showing fluorescence transients evoked using the same Poisson train across trials. Single-trial fluorescence (D) and denoised fit (calcium level plus baseline). (F) Spike detection for each trial. (G) 100 ms time window highlighting the first four stimulation-induced action potentials. Normalized fluorescence and denoised fit (top), average spike count (bottom). Orange vertical lines denote stimulation time points. (H) Comparison of the posterior distributions of the interval between the first two detected spikes across experimental trials. The solid vertical line at 5.3 ms denotes the time interval between the first two stimulations. (I) Comparison of the posterior modes of the inter-spike interval across trials between real data and simulations.

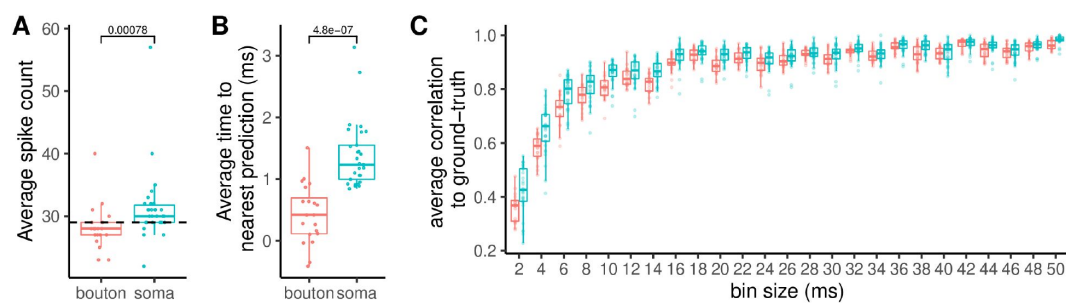


Figure 8.

Temporal resolution analysis across soma and synaptic boutons along the parallel fiber

(A) Average spike count estimated from 6 somata and 4 boutons (5 trials each). The dashed line denotes the number of stimulation time points (29). **(B)** Time from stimulus time predicted spike averaged across all 29 stimulations per trial. The comparison between somas and boutons shows that somatic transients are delayed by 1 ms. **(C)** Correlation between predicted spikes and stimulation events across time scales.

As an alternative estimate of temporal accuracy we have calculated the Pearson's correlation between detected spike times and stimulation times over different time-scales. We binned detected spike times and the Poisson stimulation train using bin sizes ranging from 2 ms to 50 ms and calculated their correlation. If we consider as reference correlation the average across the CASCADE dataset (0.75), this analysis shows that the correlation is larger than 0.75 up to 10 ms bin size, confirming that our method provides accurate inference of short intervals.

Next, we compared the temporal accuracy of PGBAR, CASCADE and MLspike predictions (**Figure 9A**). We find that the distributions of the average time from ground-truth across our experiments using MLspike, Cascade and PGBAR differ in their spread, with 10th-to-90th percentile ranges of 14 ms, 8 ms and 3 ms respectively. In addition, the false detection rate, defined as the difference between detected and ground-truth spikes divided by the ground-truth, is significantly lower in PGBAR, compared to MLspike and CASCADE (Mann-Whitney test p-value $< 10^{-3}$) which show a much broader distribution across trials (**Figure 9B**). These results demonstrate that PGBAR estimates are more reliable than MLspike and CASCADE across trials, with a narrower and unbiased distribution of both average time to nearest spike and the false positive rate. Next, we compared the correlation with ground-truth across different time scales for the three methods (**Fig. 9C**). In our cerebellar dataset, PGBAR outperforms the other methods across the whole range of bin sizes, in particular in the short time scale, highlighting further the advantage of our method for high firing rates.

To quantify how the hyperparameters used in the priors affect the PGBAR predictions, we carried out a biparametric sensitivity analysis in comparison with MLspike. For PGBAR we considered the hyperparameters of the Bayesian priors associated with the peak response to single spike and the baseline variance, which influences how much of the fluorescence can be attributed to baseline modulation. For MLspike we considered the transient amplitude and the decay time constant. For both methods we varied the parameters between -50% and +50% of their optimal value and estimated the correlation between predictions and ground-truth as well as the number of spikes (**Fig. 10A**). Both distributions of Pearson's correlation and spike number arising from different choices of parameters (model parameters for MLspike and prior hyperparameters for PGBAR) are much narrower in the case of PGBAR. To quantify this difference we calculated the coefficient of variation across all parameter configurations for each trial (**Fig. 10B**), which was found significantly lower in PGBAR compared to MLspike. Our analysis shows that PGBAR predictions have a much lower sensitivity to inaccurate choices of the prior hyperparameters compared to the stronger dependence of MLspike on its model parameters.

3 Discussion

Calcium-sensitive fluorescence indicators provide essential tools for monitoring the activity of neuronal populations in model organisms. However, the extraction of underlying firing patterns from fluorescence time series is challenging due to low signal-to-noise ratio, incomplete knowledge of the indicator dynamics, complex firing statistics, and unknown fluorescence modulation. In spite of the proliferation of methodologies developed to address this issue, limited attention has been devoted to the estimation of the statistical uncertainties associated to spike inference. The vast majority of the spike detection algorithms are indeed based on optimization techniques, providing only point estimates of the detected spikes. Quantifying statistical uncertainties is key to comparing firing patterns across neurons (Diana et al., 2019) and establishing their causal relationships. The work of Pnevmatikakis et al. (2013) and Vogelstein et al. (2009), addressed this issue by using Monte Carlo methods to approximate the full posterior distributions of spiking patterns. Building upon their work, here we improve the generative model used to infer spiking patterns from fluorescence time series and describe efficient Monte Carlo strategies to infer spike times and their statistical uncertainties.

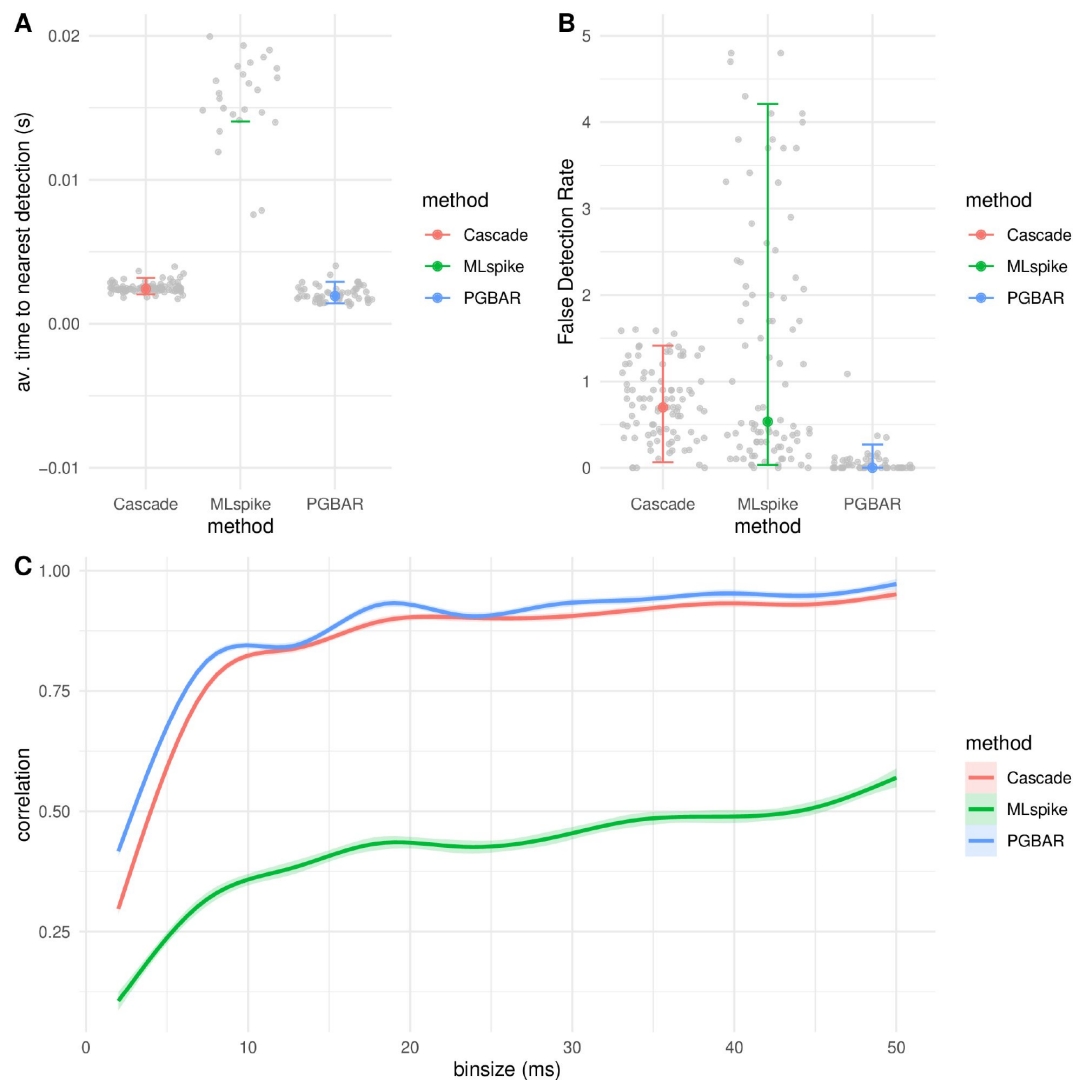


Figure 9.

Comparison of temporal accuracy across methods

(A) Estimate of temporal accuracy (absolute time difference between ground-truth spikes and nearest detections averaged across detections) for all cerebellar recordings and across inference methods (CASCADE, MLspike and PGBAR). (B) False detection rate defined as the number of excess spikes divided by the number of ground-truth spikes. (C) Pearson's correlation at different temporal scales across methods. (A,B) Error bars denote the 10th-to-90th percentile range.

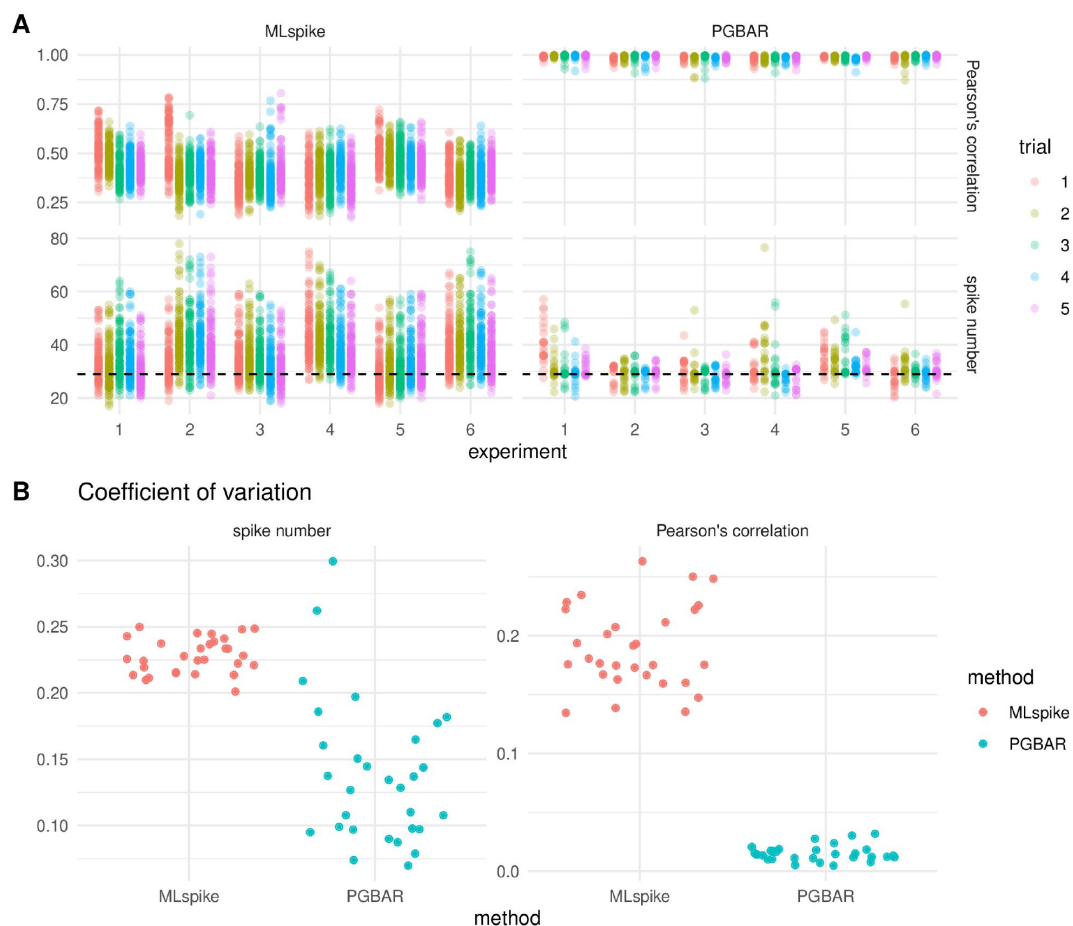


Figure 10.

Parameter sensitivity in PGBAR and MLspike

(A) Pearson's correlation and predicted spike count for PGBAR and MLspike across experiments and trials with varying PGBAR prior hyperparameters and MLspike model parameters in the range between -50% and +50% from the optimal value. (B) Coefficient of variation over parameter sets for both Pearson's correlation and spike number associated to each prediction and comparison between PGBAR and MLspike.

Bursting dynamics and baseline modulation

Neural activity patterns are not always well described by a simple Poisson spiking process. The PGBAR is the first Monte Carlo method (Greenberg et al., 2018 [↗](#); Pnevmatikakis et al., 2013 [↗](#); Vogelstein et al., 2009 [↗](#)) to perform statistical inference based on a non-homogeneous Poisson firing and baseline modulation model. PGBAR uses a two-state process to enable transitions between low and high firing rates. This feature is used to mimic the alternation between periods of low baseline firing and bursting activity transients where the firing rate is significantly increased. We have shown that not taking into account bursting activity in the model used for inference can lead to biased results, especially at low SNR levels and high firing frequency. By explicitly modeling the bursting dynamics, PGBAR produced unbiased results at all levels of noise and frequency when tested on simulated data (Figure 3 [↗](#)).

Although the model used in Pnevmatikakis et al. (2013) [↗](#) did not account for fluorescence baseline modulation, they acknowledged that it would be important to account for baseline fluctuations in *in vivo* recordings. PGBAR uses a Gaussian random walk, as MLSpoke (Deneux et al., 2016 [↗](#)), to describe slow changes in baseline fluorescence across the recording. While this is a simple Markov model of fluorescence baseline it introduces additional noise. To avoid this effect, it is possible to employ alternative baseline models, such as the integrated random walk.

Joint estimation of time-independent parameters and dynamic variables

We employed for the first time state-of-the-art particle Gibbs algorithms to infer spikes from noisy fluorescence. This is a key novelty compared to previous SMC-based methods (Greenberg et al., 2018 [↗](#); Vogelstein et al., 2009 [↗](#)), allowing for a joint estimation of time-independent parameters and dynamic variables. The estimation of time-independent model parameters is a well-known issue in spike detection algorithms, typically requiring ad-hoc calibration procedures, grid search, or manual settings. Because spike inference is sensitive to parameters such as the calcium response amplitude, rise and decay kinetics, and noise level, errors in these parameters can substantially affect the accuracy of spike time estimates. By jointly sampling model parameters and latent variables, PGBAR eliminates the need for separate calibration and ensures that uncertainty in parameters is propagated to spike time estimates in a principled way. As illustrated in Figure 10 [↗](#), this leads to a more robust inference compared to existing methods like MLSpoke, which show greater sensitivity to parameter variation. In addition, PGBAR enables the users to calibrate the inference of action potentials by setting prior mean and variance of phenomenological parameters (e.g. rise and decay constants, firing rates, bursting frequencies).

Comparison with benchmark datasets

The proliferation of spike inference methodologies led to the development of community-based initiatives (Berens et al., 2018 [↗](#); Theis et al., 2016 [↗](#)) to rank the performance of available methods. We applied our approach on the CASCADE dataset (Rupprecht et al., 2021 [↗](#)) which provides a curated database of neuronal recordings from mouse and zebrafish using different calcium indicators. The performance of PGBAR, measured by the correlation coefficient with ground-truth spikes, is within one quartile from the correlation distributions of existing unsupervised approaches. In addition, it provides information about the statistical uncertainty associated to spike detection that is not currently possible with state-of-the-art techniques.

PGBAR detection of short high-frequency bursts using an ultrafast calcium indicator

PGBAR employs a second-order autoregressive process to link spiking activity to fluorescence. This simple model accounts for the basic qualitative aspects of calcium transients, and it is well-suited for linear indicators. For this reason, we have tested the performance of PGBAR on the ultrafast GCaMP8f (Zhang et al., 2023) that exhibits improved linearity in comparison to previous calcium probes. We showed that the combination of PGBAR with the GCaMP8f enables the detection of inter-spike intervals of 5 ms with an accuracy of 2.5 ms from single trials, thus offering a statistical tool for estimating high-frequency neural activity patterns.

PGBAR limitations and future perspectives

Although full Bayesian inference is known to be computationally expensive, SMC algorithms are highly parallelizable. Posterior distributions are represented by particles that are simultaneously propagated through time. In particular, GPU parallelization of SMC methods is an active field of research in computational statistics. Future advances in this direction might dramatically boost these methods and offer tools for online processing of fluorescence time series.

Many commonly used indicators exhibit a nonlinear fluorescence response to increases in intracellular calcium (Chen et al., 2013). The autoregressive model used in this study does not account for that nonlinearity. This work provides a statistical framework that can be generalized to more specific biophysical models of calcium indicators to account for non-linear effects. In addition, these models depend on rate constants that are difficult to measure directly. Not being able to constrain model parameters is an issue for spike inference as different parameter combinations might describe equally well the observed fluorescence by adjustments of the inferred spike pattern. For this reason, when model parameters are not identifiable, the problem of spike inference from fluorescence time series is ill-defined. Our Bayesian framework offers a systematic approach to address this issue by exploiting current and future data on the kinetics of calcium indicators through informative priors. Thanks to our joint sampling strategy of model parameters and spike patterns, our Monte Carlo method can constrain the inference of model parameters to biophysically relevant ranges.

4 Materials and methods

4.1 Particle Gibbs with ancestor sampling

The PGAS step used in Algorithm 1 to sample latent state trajectories was introduced in Lindsten et al. (2014) to improve the performance of the original particle Gibbs sampler (Andrieu et al., 2010). We refer to their original works for details on convergence and mixing properties of the method. The PGAS step is a SMC algorithm that generates a new latent state trajectory starting from a reference trajectory and the model parameters. We initialize the algorithm with a set of N latent states (particles) at time $t = 1$,

$$X_1^{(i)} = \{q_1^{(i)}, s_1^{(i)}, C_1^{(i)}, b_1^{(i)}\} \quad i = 1, \dots, N \quad (14)$$

where the first $N - 1$ are sampled from a proposal distribution equal to the probability of the initial state $\mu^\theta(X_1)$

$$\mu^\theta(X_1) = \rho_1(X_1) = \frac{(r_{q_1} \Delta)^{s_1}}{s_1!} e^{-r_{q_1} \Delta} \cdot \frac{e^{-b_1^2/2}}{\sqrt{2\pi}} \delta^2(C_1 - \hat{C}_1) \quad (15)$$

where bursting and baseline firing states $q_1 = 0, 1$ have equal probability and the calcium vector is constrained by the initial condition of Eq. (5) [\[11\]](#), $\hat{C}_1 \equiv (c_0 + AS_1, 0)$ and the model parameter c_0 . The last particle is constrained by the reference trajectory X . To conclude the initialization stage, we assign importance weights $w_1^{(i)}$ to all particles

$$w_1^{(i)} = \mu^\theta(X_1^{(i)})g^\theta(F_1|X_1^{(i)})/\rho_1(X^{(i)}), \quad i = 1, \dots, N. \quad (16)$$

Next, for $t > 1$, we evolve the particle system through time by assigning ancestor particles $\{\tilde{X}_{t-1}^{(i)}\}_{i=1}^N$, propagate these to time t to get a new set of particles $\{X_{1:t}^{(i)}\}_{i=1}^N$ and assigning weights $w_t^{(i)}$. For the first $N - 1$ particles the ancestors are obtained by multinomial resampling from the particle system at time $t - 1$ with probability proportional to the importance weights $w_{t-1}^{(i)}$. The ancestor J of the last particle is drawn from the distribution

$$\mathbb{P}(J = i) = \frac{w_{t-1}^{(i)} f_t^\theta(X_t'|X_{t-1}^{(i)})}{\sum_{k=1}^N w_{t-1}^{(k)} f_t^\theta(X_t'|X_{t-1}^{(k)})} \quad (17)$$

exploiting the fact that we know its state at time t .

In order to evolve the particle system through time we need to set a proposal distribution $\rho_t(X_t|X_{t-1})$ to sample new latent states. This proposal is then taken into account in the reweighting stage. Although the choice of the proposal distribution is arbitrary, it can be shown that the conditional distribution $P(X_t|X_{t-1}, F_t)$ reduces the variance of the importance weights. We can express this optimal proposal as

$$P(X_t|F_t, X_{t-1}) = \frac{P(X_t, F_t|X_{t-1})}{\int_{X_t} P(X_t, F_t|X_{t-1})} = \frac{f_t^\theta(X_t|X_{t-1})g_t^\theta(F_t|X_t)}{Z^\theta(X_{t-1}, F_t)} \quad (18)$$

where $Z^\theta(X_{t-1}, F_t)$ is the normalization factor as a function of the latent state at time $t - 1$ and current fluorescence F_t . We can now use the expressions of f_t^θ and g_t^θ in Eqs. (11) [\[11\]](#), (12) [\[12\]](#) for our model to compute the optimal proposal distribution. To do so we decompose $P(X_t|X_{t-1}, F_t)$ as the product

$$P(X_t = \{q_t, s_t, C_t, b_t\}|X_{t-1}, F_t) = P(q_t, s_t|X_{t-1}, F_t) \cdot P(C_t|C_{t-1}, s_t) \cdot P(b_t|q_t, s_t, C_t, X_{t-1}, F_t) \quad (19)$$

where we used the fact that C_t is deterministic and only depends on C_{t-1} and the spike count at time t . The idea is to use this chain decomposition to sample first the firing state q_t and the spike count s_t , then calculate C_t from its deterministic evolution and finally sample the baseline b_t from its distribution conditional to the other variables. The first term $P(q_t, s_t|X_{t-1})$ can be obtained by integrating the product $f_t^\theta(X_t|X_{t-1})g_t^\theta(F_t|X_t)$ over b_t and C_t and then normalizing the result. The integration over b_t and C_t leads to

$$\begin{aligned} \int db_t dC_t f_t^\theta(X_t|X_{t-1})g_t^\theta(F_t|X_t) &= \int db_t dC_t \delta^{(2)}(C_t - M \cdot C_{t-1} - AS_t) \cdot W_{q_t-1q_t} \frac{(r_{q_t}\Delta)^{s_t}}{s_t!} e^{-r_{q_t}\Delta} \cdot \\ &\cdot (2\pi\Delta\sigma_b^2)^{-1/2} \exp\left(-\frac{1}{2\Delta\sigma_b^2}(b_t - b_{t-1})^2\right) \cdot (2\pi\sigma^2)^{-1/2} \cdot \exp\left[-\frac{1}{2\sigma^2}(F_t - c_t - b_t)^2\right] \\ &= W_{q_t-1q_t} \frac{(r_{q_t}\Delta)^{s_t}}{s_t!} e^{-r_{q_t}\Delta} \cdot I(b_{t-1}, F_t - c_t, \Delta\sigma_b^2, \sigma^2) \end{aligned} \quad (20)$$

where we introduced the function $I(y_1, y_2, \sigma^2, \sigma^2)$ as the integral

$$I(y_1, y_2, \sigma_1^2, \sigma_2^2) = (2\pi\sigma_1^2)^{-1/2} (2\pi\sigma_2^2)^{-1/2} \int dx e^{-\frac{(x-y_1)^2}{2\sigma_1^2} - \frac{(x-y_2)^2}{2\sigma_2^2}} = \frac{\exp\left[-\frac{1}{2} \frac{(y_1 - y_2)^2}{\sigma_1^2 + \sigma_2^2}\right]}{\sqrt{2\pi(\sigma_1^2 + \sigma_2^2)}} \quad (21)$$

The normalization factor $Z^\theta(X_{t-1}, F_t)$ is obtained by taking the sum of Eq. (20) over firing state and spike count:

$$Z^\theta(X_{t-1}, F_t) = \sum_{q' \in \{0,1\}} \sum_{s'=0}^{\infty} W_{q_{t-1}q'} \frac{(r_{q'}\Delta)^{s'}}{s'!} e^{-r_{q'}\Delta} \cdot I(b_{t-1}, F_t - c_t, \Delta\sigma_b^2, \sigma^2). \quad (22)$$

To draw a combination of q_t and s_t from this distribution we applied a cutoff to the number of spikes per time step $S^{(max)} = 20$ and constructed a probability matrix of size $2 \times S^{(max)}$ for all combinations of firing state and spike count.

To obtain the full conditional distribution of b_t we consider again the product $f_t^\theta(X_t|X_{t-1})g_t^\theta(F_t|X_t)$ and by keeping only terms in b_t and normalizing we obtained a Gaussian distribution with mean μ_{prop} and variance σ_{prop} given by

$$\mu_{prop} = \frac{b_{t-1}\sigma^2 + (F_t - c_t)\sigma_b^2\Delta}{\sigma^2 + \sigma_b^2\Delta} \quad (23)$$

$$\sigma_{prop}^2 = \left(\frac{1}{\sigma^2} + \frac{1}{\sigma_b^2\Delta} \right)^{-1}. \quad (24)$$

The final step is to reweight all particles according using the importance weight

$$w_t^{(i)} = f_t^\theta(X_t^{(i)}|\tilde{X}_{t-1}^{(i)})g_t^\theta(F_t|X_t^{(i)})/\rho_t(X_t^{(i)}|\tilde{X}_{t-1}^{(i)}) \quad (25)$$

However, due to the form of the optimal proposal in Eq. (18) the importance weights reduce to the normalization factor calculated in Eq. (22)

$$w^{(i)} = Z^\theta(X_{t-1}^{(i)}, F_t) \quad (26)$$

Algorithm 2 PGAS kernel

Input: reference trajectory $X'_{1:T}$, and model parameters θ

- 1: Draw $X_1^{(i)}$ from the poposal distribution ρ_1 for $i = 1, \dots, N - 1$
- 2: Set $X_1^{(N)} = X'_1$
- 3: Set importance weights $w_1^{(n)} = \mu^\theta(X_1^{(n)})g^\theta(F_1|X_1^{(n)})/\rho_1(X_1^{(n)})$ for $n = 1, \dots, N$
- 4: **for** t in $2:T$ **do**
// Resampling and ancestor sampling
 - 5: Resample $N - 1$ particles $\{\tilde{X}_{1:t-1}^{(i)}\}_{i=1}^{N-1}$ with probabilities proportional to the importance weights $\{w_{t-1}^{(i)}\}_{i=1}^{N-1}$
 - 6: Draw J with probability $\mathbb{P}(J = i) \propto w_{t-1}^{(i)}f_t^\theta(X_t|X_{t-1}^{(i)})$ and set $\tilde{X}_{1:t-1}^{(N)} = X_{1:t-1}^{(J)}$
// Particle propagation
 - 7: Draw $X_t^{(i)}$ from the proposal distribution $\rho_t(X_t|\tilde{X}_{t-1}^{(i)})$ for $i = 1, \dots, N - 1$
 - 8: Set $X_t^{(N)} = X'_t$
 - 9: Set $X_{1:t}^{(i)} = (\tilde{X}_{1:t-1}^{(i)}, X_t^{(i)})$ for $i = 1, \dots, N$
// Weighting
 - 10: Set $w_t^{(i)} = f_t^\theta(X_t^{(i)}|\tilde{X}_{t-1}^{(i)})g_t^\theta(F_t|X_t^{(i)})/\rho_t(X_t^{(i)}|\tilde{X}_{t-1}^{(i)})$ for $i = 1, \dots, N$
 - 11: Draw k with $\mathbb{P}(k = i) \propto w_t^{(i)}$

Output: $X_{1:T}^{(k)}$

4.2 Prior distributions

As discussed in the text, we use a reparameterization of the autoregressive model in terms of the maximal amplitude $A^{(max)}$, rise and decay times τ_r and τ_d , for which it is easier to design realistic prior distributions based on previous empirical estimates of the kinetics of calcium indicators. We have used truncated normal priors for the maximal amplitude, the initial condition of the

autoregressive model c_0 , rise and decay time. In order to calculate the full conditional distributions on bursting/baseline firing rates $r_{0,1}$ and the transition matrix parameters $w_{q \rightarrow q'}$ we have used a gamma distribution, whereas for the noise level σ^2 an inverse gamma distribution.

4.3 Sampling rules for time-independent parameters

In Algorithm 1, after a new latent state trajectory is sampled from the PGAS kernel, we draw time-independent model parameters from the conditional distribution $P(\theta_i | X_{1:T}, F_{1:T})$. We use a mixed approach where the parameters $w_{q \rightarrow q'}$ and σ^2 are sampled from their full conditional distribution, which can be obtained analytically by using gamma priors, while kernel parameters, $A^{(max)}$ and $\tau_{r,d}$ are sampled using the Metropolis-Hastings acceptance rule.

The full conditional distribution of a given parameter can be obtained from the joint probability of model parameters, latent state and fluorescence trajectories

$$P(\theta) \cdot P_\theta(X_{1:T}, F_{1:T} | \theta) \quad (27)$$

where $P(\theta)$ is the prior distribution. We will now calculate the full conditionals for firing rates r_q , transition parameters $w_{q \rightarrow q'}$ and noise variance σ . For simplicity we will use the same symbols for shape, α , and rate, β of all prior distributions, although they differ numerically for each parameter. By combining the expressions in Eqs.(10), (11) and (12) with the gamma prior $\text{gamma}(\alpha, \beta)$ and by keeping only terms proportional to $r_{0,1}$ we obtain

$$P(r_q | \dots) \propto r_q^{\alpha-1} e^{-\beta r_q - r_q \Delta T} r_q^{\sum_{t:q_t=q} s_t} \quad (28)$$

therefore the full conditional is a gamma distribution with updated parameters

$$\alpha' = \alpha + \sum_{t:q_t=q} s_t \quad (29)$$

$$\beta' = \beta + \Delta T \quad (30)$$

By applying the same method to the transition rates $w_{q \rightarrow q'}$ we have

$$P(w_{q \rightarrow q'} | \dots) \propto w_{q \rightarrow q'}^{\alpha-1} e^{-\beta w_{q \rightarrow q'}} w_{q \rightarrow q'}^{N_{qq'}} (1 - \Delta w_{q \rightarrow q'})^{N_{qq}} \approx w_{q \rightarrow q'}^{\alpha+N_{qq'}-1} e^{-(\beta+\Delta N_{qq})w_{q \rightarrow q'}} \quad (31)$$

where the approximation holds when the transition rate between firing states is much slower than the sampling frequency ($w_{q \rightarrow q'} \ll \Delta^{-1}$). Therefore the full conditional is again a gamma distribution with parameters

$$\alpha' = \alpha + N_{qq'} \quad (32)$$

$$\beta' = \beta + \Delta N_{qq} \quad (33)$$

For the noise variance parameter we used an inverse gamma prior and by applying the same method we can compute the full conditional as

$$P(\sigma^2 | \dots) \propto (\sigma^2)^{-\alpha-1-T/2} \exp \left(-\frac{\beta}{\sigma^2} - \frac{1}{2\sigma^2} \sum_t (F_t - c_t - b_t)^2 \right) \quad (34)$$

therefore the updated shape and rate of the inverse gamma are

$$\alpha' = \alpha + T/2 \quad (35)$$

$$\beta' = \beta + \frac{1}{2} \sum_t (F_t - c_t - b_t)^2 \quad (36)$$

4.4 Response kernel

The response to a single spike can be obtained by writing Eq. (3) in the form of a first order Markov process in terms of the new variables $C_t \equiv [c_t, c_{t-1}]$ and $S_t \equiv [s_t, 0]$ so that

$$C_t = M \cdot C_{t-1} + AS_t, \quad M = \begin{bmatrix} \gamma_1 & \gamma_2 \\ 1 & 0 \end{bmatrix} \quad (37)$$

with the initial condition $C_1 = [c_0 + AS_1, 0]$. We can now write the solution at time t as

$$C_t = M^{t-1} C_1 + A \sum_{k=2}^t M^{t-k} S_k. \quad (38)$$

If $s_t = \delta_{t,1}$ and $c_0 = 0$ then Eq. (38) simplifies to

$$C_t = AM^{t-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \quad (39)$$

By introducing eigenvectors and eigenvalues of M

$$\gamma_{\pm} \equiv \frac{\gamma_1 \pm \sqrt{\gamma_1^2 + 4\gamma_2}}{2}, \quad e_{\pm} \equiv \begin{bmatrix} \gamma_{\pm} \\ 1 \end{bmatrix} \quad (40)$$

we can express Eq. (39) as

$$C_t = AM^{t-1} \left(\frac{e_+ - e_-}{\gamma_+ - \gamma_-} \right) = A \left(\frac{\gamma_+^{t-1} e_+ - \gamma_-^{t-1} e_-}{\gamma_+ - \gamma_-} \right) \quad (41)$$

therefore we have

$$c_t = A \left(\frac{\gamma_+^t - \gamma_-^t}{\gamma_+ - \gamma_-} \right). \quad (42)$$

By setting the time derivative of c_t to zero we obtain the time to reach the maximal response τ_r as

$$\tau_r = \frac{\log \left(\frac{g_+}{g_-} \right)}{g_- - g_+}, \quad g_{\pm} = \log \gamma_{\pm} \quad (43)$$

whereas if we take the long time limit of Eq. (39) we obtain

$$c_t = A\gamma_+^t [1 - (\gamma_-/\gamma_+)^t] \approx Ae^{t/\tau_d}, \quad \tau_d = -\frac{1}{g_+} \quad (44)$$

4.5 Reparameterization

To reparameterize the autoregressive model in terms of kinetic parameters we need to find the inverse map $\tau_{r,d} \rightarrow \gamma_{1,2}$ to obtain the original autoregressive parameters given rise and decay times. By combining the expressions of τ_r and τ_d in terms of $g_{\pm}$ we have

$$\frac{\tau_r}{\tau_d} = \frac{\log \frac{g_-}{g_+}}{\frac{g_-}{g_+} - 1} \quad (45)$$

which shows that the ratio g_-/g_+ can be expressed as

$$\frac{g_-}{g_+} = f^{-1}\left(\frac{\tau_r}{\tau_d}\right), \quad f(x) = \frac{\log(x)}{x-1} \quad (46)$$

where the inverse function $f^{-1}(x)$ can be determined by numerical interpolation in the range [0,1]. To obtain the original autoregressive parameters $\gamma_{1,2}$ we first obtain $g_{\pm}$ as

$$g_+ = -\frac{1}{\tau_d} \quad (47)$$

$$g_- = g_+ \cdot f^{-1}\left(\frac{\tau_r}{\tau_d}\right) \quad (48)$$

and then

$$\gamma_1 = e^{g_+} + e^{g_-} \quad (49)$$

$$\gamma_2 = -e^{g_+ + g_-} \quad (50)$$

4.6 Runtime benchmark

We conducted a thorough runtime benchmark analysis to evaluate the speed of PGBAR under different number of particles used in the PGAS kernel and the length of the time series to analyze. Specifically we measured the execution time of a single iteration of [Algorithm 1](#) to draw a sample of time-dependent variables and the model parameters. The runtime was averaged over the generation of 100 posterior samples. We performed this test by varying the number of particles from 50 to 1000, using time series lengths ranging from 200 to 1000 time steps (**Fig. 11**). The time to draw a single iteration of PGBAR ranges from a minimum of 40 ms, to process a time series of 200 time steps with 50 PGAS particles, up to 2.3 s to process 1000 time steps with 1000 particles. Our analysis demonstrates that the runtime scales linearly both in the number of PGAS particles and the time series length. All tests were executed on a Dell XPS 15 laptop with with CPU 11th Gen Intel(R) Core(TM) i7-11800H @ 2.30GHz and 32Gb of RAM.

4.7 Experimental methods

GCaMP8f virus injection in the Crus I lobule of the cerebellum

Virus injection was targeted to Crus I (6.00 mm posterior to the Bregma, 3.00 mm lateral to the midline; vertical depth of 0.50 mm from pial surface) under deep isofluorane anesthesia. 100 nl of adeno-associated virus encoding GCaMP8f (AAV-DJ-CAG-FLEX-jGCaMP8f, Janelia Research Campus) was injected with Nanoject III (Drummond Scientific) using thin glass pipette (diameter 30 μ m). Injection was made when the C57BL/6J-Gabra6tm2(cre)Wwis mice were 7 weeks old. After injections the mice were returned to their home cage for 6 weeks to allow time for expression.

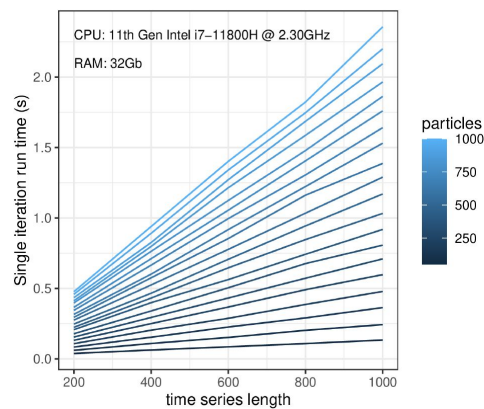


Figure 11.

Runtime benchmarking. Execution time of a single iteration of Algorithm 1 sampling both model parameters and time-dependent variables. The performance was estimated by averaging over 100 iterations. All tests were conducted on a XPS 15 Dell laptop with CPU 11th Gen Intel(R) Core(TM) i7-11800H @ 2.30GHz and 32Gb of RAM.

Slice Preparation

Acute coronal slices (200 μm) of the Crus I lobule of the cerebellum was prepared from adult C57BL/6J-Gabra6^{tm2(cre)}W^{wis} mice, aged 97 and 102 days. Following transcardial perfusion with an ice-cold solution containing (in mM): 2.5 KCl, 0.5 CaCl₂, 4 MgCl₂, 1.25 NaH₂PO₄, 24 NaHCO₃, 25 glucose, 230 sucrose, and 0.5 ascorbic acid, the brains were removed and placed in the same solution. The solution was bubbled with 95% O₂ and 5% CO₂. Slices were cut from the dissected lateral cerebellum using a vibratome (Leica VT1200S), and incubated at room temperature for 30 minutes in a solution containing (in mM): 85 NaCl, 2.5 KCl, 0.5 CaCl₂, 4 MgCl₂, 1.25 NaH₂PO₄, 24 NaHCO₃, 25 glucose, 75 sucrose and 0.5 ascorbic acid. Slices were then transferred to an external recording solution containing (in mM): 125 NaCl, 2.5 KCl, 1.5 CaCl₂, 1.5 MgCl₂, 1.25 NaH₂PO₄, 24 NaHCO₃, 25 glucose and 0.5 ascorbic acid, and maintained at room temperature for up to 6 hr.

Cellular Imaging

The brain slices containing GCaMP8f expressing cells were identified with a 4x objective lens (Olympus UplanFI 4x, 0.13 NA) using very brief illumination with 470 nm light to excite GCaMP8f fluorescence. GCs were identified using infrared Dodt-gradient contrast and a QIClick digital CCD camera (QImaging, Surrey, BC, Canada) mounted on an Ultima multiphoton microscopy system (Bruker Nano Surfaces Division, Middleton, WI, USA) that was mounted on an Olympus BX61W1 microscope, equipped with a water-immersion objective (Olympus 60x, 1.1 NA). Two-photon excitation was performed with a Ti-sapphire laser (Spectraphysics). To visualize GCs expressing GCaMP8f, two-photon excitation was performed at 920 nm. Infrared Dodt-gradient contrast was used to position the stimulation pipette in molecular layer targeting PF to directly activate GC bodies. Linescan imaging of GC bodies was performed by scanning through the whole cell membrane marked by freehand linescan mode (Prairie View). Total laser illumination per sweep lasted 2000 ms. Fluorescence was detected using both proximal epifluorescence and substage photomultiplier tube gallium arsenide phosphide (H7422PA-40 SEL, Hamamatsu).

Code and Data

In this work we employed the publicly available CASCADE dataset in [Rupprecht et al. \(2021\)](#). Linescan imaging data and the source code of PGBAR are available through the GitHub repository <https://github.com/giovannidiana/pgbar>.

Acknowledgements

This project has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement No 896051. This work was also supported by a Pasteur-Roux-Cantarini fellowship of the Institut Pasteur. We would like to thank Peter Rupprecht for his help with the analysis of the CASCADE dataset and for thorough discussions. We also thank Nicolas Chopin for suggesting the use of backward steps methods in particle Gibbs, Andrea Giovannucci, Marco Banterle, Diana Passaro and Fredrik Lindsten for helpful discussions.

References

- Andrieu C, Doucet A, Holenstein R. (2010) **Particle markov chain monte carlo methods** *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* **72**:269–342 [Google Scholar](#)
- Berens P, Freeman J, Deneux T, Cherkov N, McColgan T, Speiser A, Macke JH, Turaga SC, Mineault P, Rupprecht P, et al. (2018) **Community-based benchmarking improves spike rate inference from two-photon calcium imaging data** *PLoS computational biology* **14**:e1006157 [Google Scholar](#)
- Chen TW, Wardill TJ, Sun Y, Pulver SR, Renninger SL, Baohan A, Schreiter ER, Kerr RA, Orger MB, Jayaraman V, et al. (2013) **Ultrasensitive fluorescent proteins for imaging neuronal activity** *Nature* **499**:295–300 [Google Scholar](#)
- Chopin N, Papaspiliopoulos O. (2020) **An Introduction to Sequential Monte Carlo** Springer-Verlag [Google Scholar](#)
- Deneux T, Kaszas A, Szalay G, Katona G, Lakner T, Grinvald A, Rózsa B, Vanzetta I. (2016) **Accurate spike estimation from noisy calcium signals for ultrafast three-dimensional imaging of large neuronal populations in vivo** *Nature communications* **7**:1–17 [Google Scholar](#)
- Diana G, Sainsbury TT, Meyer MP (2019) **Bayesian inference of neuronal assemblies** *PLoS computational biology* **15**:e1007481 [Google Scholar](#)
- Dyer EL, Studer C, Robinson JT, Baraniuk RG (2013) **A robust and efficient method to recover neural events from noisy and corrupted data** In: *2013 6th International IEEE/EMBS Conference on Neural Engineering (NER)* IEEE pp. 593–596 [Google Scholar](#)
- Fletcher AK, Rangan S. (2014) **Scalable inference for neuronal connectivity from calcium imaging** *arXiv* [Google Scholar](#)
- Friedrich J, Paninski L. (2016) **Fast active set methods for online spike inference from calcium imaging** *Advances In Neural Information Processing Systems* **29**:1984–1992 [Google Scholar](#)
- Friedrich J, Zhou P, Paninski L. (2017) **Fast online deconvolution of calcium imaging data** *PLoS computational biology* **13**:e1005423 [Google Scholar](#)
- Giovannucci A, Friedrich J, Gunn P, Kalfon J, Brown BL, Koay SA, Taxidis J, Najafi F, Gauthier JL, Zhou P, et al. (2019) **CaImAn an open source tool for scalable calcium imaging data analysis** *eLife* **8**:e38173 <https://doi.org/10.7554/eLife.38173> | [Google Scholar](#)
- Greenberg DS, Wallace DJ, Voit KM, Wuertenberger S, Czubayko U, Monsees A, Handa T, Vogelstein JT, Seifert R, Groemping Y, et al. (2018) **Accurate action potential inference from a calcium sensor protein through biophysical modeling** *BioRxiv* :479055 [Google Scholar](#)
- Grewe BF, Langer D, Kasper H, Kampa BM, Helmchen F. (2010) **High-speed in vivo calcium imaging reveals neuronal network activity with near-millisecond precision** *Nature*

methods **7**:399–405 [Google Scholar](#)

Hoang H, Ma Sato, Shinomoto S, Tsutsumi S, Hashizume M, Ishikawa T, Kano M, Ikegaya Y, Kitamura K, Kawato M, et al. (2020) **Improved hyperacuity estimation of spike timing from calcium imaging** *Scientific reports* **10**:1–16 [Google Scholar](#)

Holekamp TF, Turaga D, Holy TE (2008) **Fast three-dimensional fluorescence imaging of activity in neural populations by objective-coupled planar illumination microscopy** *Neuron* **57**:661–672 [Google Scholar](#)

Huys QJ, Paninski L. (2009) **Smoothing of, and parameter estimation from, noisy biophysical recordings** *PLoS computational biology* **5**:e1000379 [Google Scholar](#)

Im DJ, Prakhya S, Yan J, Turaga S, Branson K. (2019) **Importance Weighted Adversarial Variational Autoencoders for Spike Inference from calcium Imaging Data** *arXiv* [Google Scholar](#)

Jewell S, Witten D. (2018) **Exact spike train inference via ℓ_0 optimization** *The annals of applied statistics* **12**:2457 [Google Scholar](#)

Jewell SW, Hocking TD, Fearnhead P, Witten DM (2020) **Fast nonconvex deconvolution of calcium imaging data** *Biostatistics* **21**:709–726 [Google Scholar](#)

Kazemipour A, Liu J, Solarana K, Nagode DA, Kanold PO, Wu M, Babadi B. (2017) **Fast and stable signal deconvolution via compressible state-space models** *IEEE Transactions on Biomedical Engineering* **65**:74–86 [Google Scholar](#)

Kerr JN, Greenberg D, Helmchen F. (2005) **Imaging input and output of neocortical networks in vivo** *Proceedings of the National Academy of Sciences* **102**:14063–14068 [Google Scholar](#)

Lindsten F, Jordan MI, Schon TB (2014) **Particle Gibbs with ancestor sampling** *Journal of Machine Learning Research* **15**:2145–2184 [Google Scholar](#)

Lütcke H, Gerhard F, Zenke F, Gerstner W, Helmchen F. (2013) **Inference of neuronal network spike dynamics and topology from calcium imaging data** *Frontiers in neural circuits* **7**:201 [Google Scholar](#)

Malik WQ, Schummers J, Sur M, Brown EN (2011) **Denoising two-photon calcium imaging data** *PloS one* **6**:e20490 [Google Scholar](#)

Mishchenko Y, Vogelstein JT, Paninski L. (2011) **A Bayesian approach for inferring neuronal connectivity from calcium fluorescent imaging data** *The Annals of Applied Statistics* :1229–1261 [Google Scholar](#)

Mishchenko Y, Paninski L. (2011) **Efficient methods for sampling spike trains in networks of coupled neurons** *The Annals of Applied Statistics* :1893–1919 [Google Scholar](#)

Mukamel EA, Nimmerjahn A, Schnitzer MJ (2009) **Automated analysis of cellular signals from large-scale calcium imaging data** *Neuron* **63**:747–760 [Google Scholar](#)

- Oñativia J, Dragotti PL (2015) **Sparse sampling: theory, methods and an application in neuroscience** *Biological cybernetics* **109**:125–139 [Google Scholar](#)
- Oñativia J, Schultz SR, Dragotti PL (2013) **A finite rate of innovation algorithm for fast and accurate spike detection from two-photon calcium imaging** *Journal of neural engineering* **10**:046017 [Google Scholar](#)
- Pachitariu M, Stringer C, Dipoppa M, Schröder S, Rossi LF, Dalglish H, Carandini M, Harris KD (2017) **Suite2p: beyond 10,000 neurons with standard two-photon microscopy** *BioRxiv* [Google Scholar](#)
- Pnevmatikakis EA, Gao Y, Soudry D, Pfau D, Lacefield C, Poskanzer K, Bruno R, Yuste R, Paninski L. (2014) **A structured matrix factorization framework for large scale calcium imaging data analysis** *arXiv* [Google Scholar](#)
- Pnevmatikakis EA, Merel J, Pakman A, Paninski L. (2013) **Bayesian spike inference from calcium imaging data** In: *2013 Asilomar Conference on Signals, Systems and Computers IEEE* pp. 349–353 [Google Scholar](#)
- Pnevmatikakis EA, Soudry D, Gao Y, Machado TA, Merel J, Pfau D, Reardon T, Mu Y, Lacefield C, Yang W, et al. (2016) **Simultaneous denoising, deconvolution, and demixing of calcium imaging data** *Neuron* **89**:285–299 [Google Scholar](#)
- Quan T, Liu X, Lv X, Chen WR, Zeng S. (2010) **Method to reconstruct neuronal action potential train from two-photon calcium imaging** *Journal of biomedical optics* **15**:066002 [Google Scholar](#)
- Rahmati V, Kirmse K, Marković D, Holthoff K, Kiebel SJ (2016) **Inferring neuronal dynamics from calcium imaging data using biophysical models and Bayesian inference** *PLoS computational biology* **12**:e1004736 [Google Scholar](#)
- Ranganathan GN, Koester HJ (2010) **Optical recording of neuronal spiking activity from unbiased populations of neurons with high spike detection efficiency and high temporal precision** *Journal of neurophysiology* **104**:1812–1824 [Google Scholar](#)
- Rupprecht P, Carta S, Hoffmann A, Echizen M, Blot A, Kwan AC, Dan Y, Hofer SB, Kitamura K, Helmchen F, et al. (2021) **A database and deep learning toolbox for noise-optimized, generalized spike inference from calcium imaging** *Nature Neuroscience* **24**:1324–1337 [Google Scholar](#)
- Sasaki T, Takahashi N, Matsuki N, Ikegaya Y. (2008) **Fast and accurate detection of action potentials from somatic calcium fluctuations** *Journal of neurophysiology* **100**:1668–1676 [Google Scholar](#)
- Sebastian J, Kumar MG, Viraraghavan VS, Sur M, Murthy HA (2019) **Spike Estimation From Fluorescence Signals Using High-Resolution Property of Group Delay** *IEEE Transactions on Signal Processing* **67**:2923–2936 [Google Scholar](#)
- Sebastian J, Sur M, Murthy HA, Magimai-Doss M. (2021) **Signal-to-signal neural networks for improved spike estimation from calcium imaging data** *PLoS Computational Biology* **17**:e1007921 [Google Scholar](#)
- Shibue R, Komaki F. (2020) **Deconvolution of calcium imaging data using marked point processes** *PLoS computational biology* **16**:e1007650 [Google Scholar](#)

Speiser A, Yan J, Archer E, Buesing L, Turaga SC, Macke JH (2017) **Fast amortized inference of neural activity from calcium imaging data with variational autoencoders** *arXiv* [Google Scholar](#)

Stern M, Shea-Brown E, Witten D. (2020) **Inferring the spiking rate of a population of neurons from wide-field calcium imaging** *bioRxiv* [Google Scholar](#)

Theis L, Berens P, Froudarakis E, Reimer J, Rosón MR, Baden T, Euler T, Tolias AS, Bethge M. (2016) **Benchmarking spike rate inference in population calcium imaging** *Neuron* **90**:471–482 [Google Scholar](#)

Theis L, Chagas AM, Arnstein D, Schwarz C, Bethge M. (2013) **Beyond GLMs: a generative mixture modeling approach to neural system identification** *PLoS computational biology* **9**:e1003356 [Google Scholar](#)

Tsunoda T, Oda Y, Omori T, Okada M, Inoue M, Miyakawa H, Aonishi T. (2010a) **Statistical Calibration Method for Physiological Ca²⁺ Fluorescence Signals** *Australian Journal of Intelligent Information Processing Systems* **11** [Google Scholar](#)

Tsunoda T, Omori T, Miyakawa H, Okada M, Aonishi T. (2010b) **Estimation of intracellular calcium ion concentration by nonlinear state space modeling and expectation-maximization algorithm for parameter estimation** *Journal of the Physical Society of Japan* **79**:124801 [Google Scholar](#)

Tubiana J, Wolf S, Panier T, Debregeas G. (2020) **Blind deconvolution for spike inference from fluorescence recordings** *Journal of Neuroscience Methods* **342**:108763 [Google Scholar](#)

Vogelstein JT, Packer AM, Machado TA, Sippy T, Babadi B, Yuste R, Paninski L. (2010) **Fast nonnegative deconvolution for spike train inference from population calcium imaging** *Journal of neurophysiology* **104**:3691–3704 [Google Scholar](#)

Vogelstein JT, Watson BO, Packer AM, Yuste R, Jedynak B, Paninski L. (2009) **Spike inference from calcium imaging using sequential Monte Carlo methods** *Biophysical journal* **97**:636–655 [Google Scholar](#)

Wei XX, Zhou D, Grosmark A, Ajabi Z, Sparks F, Zhou P, Brandon M, Losonczy A, Paninski L. (2020) **A zero-inflated gamma model for deconvolved calcium imaging traces** *arXiv* [Google Scholar](#)

Yaksi E, Friedrich RW (2006) **Reconstruction of firing rate changes across neuronal populations by temporally deconvolved Ca²⁺ imaging** *Nature methods* **3**:377–383 [Google Scholar](#)

Zhang Y, Rózsa M, Liang Y, Bushey D, Wei Z, Zheng J, Reep D, Broussard GJ, Tsang A, Tsegaye G, et al. (2023) **Fast and sensitive GCaMP calcium indicators for imaging neural populations** *Nature* **615**:884–891 [Google Scholar](#)

Zhou P, Resendez SL, Rodriguez-Romaguera J, Jimenez JC, Neufeld SQ, Giovannucci A, Friedrich J, Pnevmatikakis EA, Stuber GD, Hen R, et al. (2018) **Efficient and accurate extraction of in vivo calcium signals from microendoscopic video data** *eLife* **7**:e28728 <https://doi.org/10.7554/eLife.28728> | [Google Scholar](#)

Author information

Giovanni Diana

Institut Pasteur, University of Paris, CNRS UMR 3571, Synapse and Circuit Dynamics
Laboratory, Paris, France
ORCID iD: [0000-0001-7497-5271](https://orcid.org/0000-0001-7497-5271)

For correspondence: g.diana.mail@gmail.com

B Semihcan Sermet

Institut Pasteur, University of Paris, CNRS UMR 3571, Synapse and Circuit Dynamics
Laboratory, Paris, France

Gerard J Broussard

Princeton Neuroscience Institute, Princeton, United States

Samuel S.-H Wang

Princeton Neuroscience Institute, Princeton, United States

David A DiGregorio

Institut Pasteur, University of Paris, CNRS UMR 3571, Synapse and Circuit Dynamics
Laboratory, Paris, France, Department of Physiology and Biophysics, University of Colorado
School of Medicine, Aurora, United States
ORCID iD: [0000-0002-6417-4566](https://orcid.org/0000-0002-6417-4566)

For correspondence: david.digregorio@cuanschutz.edu

Editors

Reviewing Editor

Lisa Giocomo

Stanford School of Medicine, Stanford, United States of America

Senior Editor

Panayiota Poirazi

FORTH Institute of Molecular Biology and Biotechnology, Heraklion, Greece

Reviewer #1 (Public review):

Summary:

In this study, Diana et al. present a Monte Carlo-based method to perform spike inference from calcium imaging data. A particular strength of their approach is that they can estimate not only averages but also uncertainties of the modeled process. The authors focus on the quantification of spike time uncertainties in simulated data and in data recorded with high sampling rate in cerebellar slices with GCaMP8f, and they demonstrate the high temporal precision that can be achieved with their method to estimate spike timing.

Strengths:

- The author provide a solid ground work for sequential Monte Carlo-based spike inference, which extends previous work of Pnevmatikakis et al., Greenberg et al. and others.

- The integration of two states (silence vs. burst firing) seems to improve the performance of the model.

- The acquisition of a GCaMP8f dataset in cerebellum is useful and helps make the point that high spike time inference precision is possible under certain conditions.

Weaknesses:

- Although the algorithm is compared (in the revised manuscript) to other models to infer individual spikes (e.g., MLSpike), these comparisons could be more comprehensive. Future work that benchmarks this and other algorithms under varying conditions (e.g., noise levels, temporal resolution, calcium indicators) would help assess and confirm robustness and useability of this algorithm.

- The mathematical complexity underlying the method may pose challenges for experimentalist who may want to use the methods for their analyses. While this is not a weakness of the approach itself, this highlights the need for further validation and benchmarking in future work, to build user confidence.

<https://doi.org/10.7554/eLife.94723.2.sa2>

Reviewer #2 (Public review):

Summary:

Methods to infer action potentials from fluorescence-based measurements of intracellular calcium dynamics are important for optical measurements of activity across large populations of neurons. The variety of existing methods can be separated into two broad classes: a) model-independent approaches that are trained on ground truth datasets (e.g., deep networks), and b) approaches based on a model of the processes that link action potentials to calcium signals. Models usually contains parameters describing biophysical variables, such as rate constants of the calcium dynamics and features of the calcium indicator. The method presented here, PGBAR, is model-based and uses a Bayesian approach. A novelty of PGBAR is that static parameters and state variables are jointly estimated using particle Gibbs sampling, a sequential Monte Carlo technique that can efficiently sample the latent embedding space.

Strengths:

A main strength of PGBAR is that it provides probability distributions rather than point estimates of spike times. This is different from most other methods and may be an important feature in cases when estimates of uncertainty are desired. Another important feature of PGBAR is that it estimates not only the state variable representing spiking activity, but also other variables such as baseline fluctuations and stationary model variables, in a joint process. PGBAR can therefore provide more information than various other methods. The information in the github repository is well-organized.

Weaknesses:

On the other hand, the accuracy of spike train reconstructions is not higher than that of other model-based approaches, and clearly lower than the accuracy of a model-independent approach based on a deep network. The authors demonstrate convincingly that PGBAR can resolve inter-spike intervals in the range of 5 ms using fluorescence data obtained with a very

fast genetically encoded calcium indicator at very high sampling rates (line scans at ≥ 1 kHz).

<https://doi.org/10.7554/eLife.94723.2.sa1>

Author response:

The following is the authors' response to the previous reviews

We thank the Reviewers and the Editor for their thoughtful and constructive feedback. In the revised manuscript, we have addressed all comments thoroughly and made several substantial improvements:

- **Benchmarking against state-of-the-art methods:** We now provide a detailed comparison of our method, PGBAR, with MLspike and CASCADE using our cerebellar dataset recorded at high sampling rates. This comparison demonstrates that PGBAR offers more reliable spike time estimates with significantly lower variability in temporal accuracy (Figure 9).
- **Quantitative analyses:** We replaced qualitative statements with quantitative metrics. For example, we now report Pearson's correlation (>0.95) of spike probabilities across trials and 100% of posterior samples with correct spike number detection during low SNR conditions (Figures 7 and 8).
- **Clarified modeling rationale:** We elaborated on the motivation behind modeling bursting dynamics using a hidden two-state process, which helps mitigate bias in spike detection under non-stationary firing conditions.
- **Model identifiability and robustness:** We demonstrate that our approach avoids parameter degeneracy through careful model design and parameter reparameterization. Sensitivity analyses (Figure 10) show that PGBAR is more robust to hyperparameter variation than MLspike.
- **Improved clarity and accessibility:** We revised the Introduction and Results sections to better explain the context, goals, and implications of our method, and clarified the advantages of joint parameter and state inference within our Bayesian framework.

We believe that these additions significantly strengthen our manuscript and demonstrate the utility of PGBAR for high-temporal-precision spike inference. Please find below our detailed responses to both Public Reviews and Recommendations for the authors.

Public Reviews

Reviewer #1 (Public Review):

Summary:

In this study, Diana et al. present a Monte Carlo-based method to perform spike inference from calcium imaging data. A particular strength of their approach is that they can estimate not only averages but also uncertainties of the modelled process. The authors then focus on the quantification of spike time uncertainties in simulated data and in data recorded with a high sampling rate in cerebellar slices with GCaMP8f.

Strengths:

- The authors provide a solid groundwork for sequential Monte Carlo-based spike inference, which extends previous work of Pnevmatikakis et al., Greenberg et al., and others.

- The integration of two states (silence vs. burst firing) seems to improve the performance of the model.

- The acquisition of a GCaMP8f dataset in the cerebellum is useful and helps make the point that high spike time inference precision is possible under certain conditions.

Weaknesses:

- The algorithm is designed to predict single spike times. Currently, it is not benchmarked against other algorithms in terms of single spike precision and spike time errors. A benchmarking with the most recent other SMC model and another good model focused on single spike outputs (e.g., MLSpikes) would be useful to have.

We thank the reviewer for the observation. In our revised manuscript, we have included a detailed comparison of spike time accuracy between our method, MLspike, and the supervised method, CASCADE, now summarized in Figure 9. In this analysis, we used our *in vitro* dataset to estimate the average temporal accuracy of spike detection across the three methods. As discussed in the main text, the average temporal accuracy was defined as the time difference between ground truth and the nearest detected spikes averaged across the ground truth. The distributions of temporal accuracies across our experiments obtained from MLspike, Cascade, and PGBAR differ in their spread, with 10th-to-90th percentile ranges of 14 ms, 8 ms, and 3 ms, respectively. This result demonstrates that PGBAR spike time estimates are more reliable than MLspike and CASCADE across trials, with a narrower unbiased distribution of temporal accuracy.

A direct comparison of PGBAR with the Sequential Binding Model (SBM) developed by Greenberg et al. was not possible since the biophysical model is designed around early GCaMP variants and thus not suitable for inference with our GCaMP8f dataset. We generally agree that employing realistic models of the calcium indicator can improve inference, however, PGBAR responds to a different question, namely how to simultaneously infer spike times and model parameters, which was still an issue with the SBM approach.

Some of the analyses and benchmarks seem too cursory, and the reporting simply consists of a visual impression of results instead of proper analysis and quantification. For example, the authors write "The spike patterns obtained using our method are very similar across trials, showing that PGBAR can reliably detect single-trial action potential-evoked GCaMP8f fluorescence transients." This is a highly qualitative statement, just based on the (subjective) visual impression of a plot. Similarly, the authors write "we could reliably identify the two spikes in each trial", but this claim is not supported by quantification or a figure, as far as I can see.

We thank the reviewer for this remark. We have now justified quantitatively our statement regarding the similarity across trials. In the revised preprint, we explain that in the specific experiment illustrated in Figure 7, Pearson's pairwise correlation between spike probabilities (Gaussian filtered with 20 ms bandwidth) across trials is always larger than 0.95. The statement quoted by the reviewer, "we could reliably identify the two spikes in each trial" refers to the fact that in 100% of the posterior samples, generated from the analysis of each trial, we detected 2 spikes in the time window considered. The temporal accuracy of our detection was then illustrated for all trials in Figure 7H, where we compared the posterior distribution of the inter-spike interval between the first two spikes across trials.

The statement referred by the Reviewer has been revised to read

(line 319) "The Pearson's pairwise correlation between spike probabilities (Gaussian filtered with 20 ms bandwidth) across trials is always larger than 0.95, which demonstrates that

PGBAR provides robust predictions across trials and it can reliably detect single-trial action potential-evoked GCaMP8f fluorescence transients.”

We revised the second statement as:

(line 324) “Despite the relatively low SNR, 100% of the posterior samples contained two spikes in the considered time interval.”

The authors write "but the trade-off between temporal accuracy, SNR and sampling frequency must be considered", but they don't discuss these trade-offs systematically.

We thank the reviewer for the comment. We have now removed the quoted sentence in the updated preprint. We revised this statement to read:

(line 302) “Based on this analysis we expect PGBAR to provide accurate estimates of inter-spike intervals down to 5 ms.”

It has been shown several times from experimental data that spike inference with single spike resolution does not work well (Huang et al. eLife, 2021; Rupprecht et al., Nature Neuroscience, 2021) in general. This limitation should be discussed with respect to the applicability of the proposed algorithm for standard population calcium imaging data.

We thank the reviewer for this comment. Detecting single spike times is indeed a difficult task. Compared to previous methods for single spike estimation, the advantage of our statistical approach is the rigorous analysis of uncertainties propagated by unknown model parameters and noisy recordings. This is an important aspect that was missing in previous approaches and that we were able to address thanks to our fully probabilistic approach.

Several analyses are based on artificial, simulated data with simplifying assumptions. Ever since Theis et al., Neuron, 2016, it has been known that artificially generated ground truth data should not be used as the primary means to evaluate spike inference algorithms. It would have been informative if the authors had used either the CASCADE dataset or their cerebellum dataset for more detailed analyses, in particular of single spike time precision.

We thank the reviewer for this comment.

To address the reviewer’s concern about single spike time precision, we have added to our revised preprint a further comparison between the temporal accuracy of PGBAR, CASCADE, and MLspike for our cerebellar dataset (Fig. 9, already discussed above).

Nevertheless, as pointed out by the reviewer, simulated data should not be used as the primary means to evaluate the performance of an inference algorithm. However, it is standard practice in the field of model-based inference to validate the approach first with data generated by the same model used for inference. This step is usually done for two main reasons: first, for internal consistency of the method, and second, to explore the regimes where inference is achievable. We made use of simulated data to address specific questions. Specifically, in Figure 2, we illustrate the analysis of data simulated using the same model for inference. In Figure 3, we used simulated data to highlight the importance of modeling bursting activity to avoid biases induced by non-homogeneous firing rates. In Figure 6, we used simulated data to explore the theoretical accuracy of PGBAR under different conditions of signal-to-noise ratio and acquisition frequencies.

In its current state, the sum of the current weaknesses makes the suggested method, while interesting for experts, rather unattractive for experimentalists who want to perform spike inference on their recorded calcium imaging data.

In our preprint, we illustrated the application of PGBAR to benchmark data and our cerebellar recordings. Therefore, our approach can be part of the calcium imaging data analysis pipeline. The advantage of estimating statistical uncertainties and model parameters makes PGBAR an attractive tool for the wide neuroscience community interested in spike inference and statistical accuracy. In addition, as noted by Reviewer 2, our code is well documented. User-friendliness and integrating our method within GUI analysis software might be the next step if there is increasing interest in using this method.

Other comments:

One of the key features of the SMC model is the assumption of two states (bursting vs. non-bursting). However, while it seems clear that this approach is helpful, it is not clear where this idea comes from, from an observation of the data or another concept.

We thank the reviewer for this comment. As the reviewer pointed out, accounting for two firing regimes is helpful as it prevents biases in estimating the number of spikes when the firing rate is non-stationary and does not follow single-frequency Poisson statistics (as shown in Figure 3 of our preprint), as expected during in vivo recordings. Animals can alter their behavioral state and be exposed to different sensory stimulations, which condition the activity of neurons. A first step beyond the assumption of a steady firing rate is indeed to introduce a hidden two-state process to separate periods of high and low firing rates. In our revised text, we explicitly discuss the rationale behind this choice. We want to emphasize that PGBAR is the only model-based approach that accounts for nonhomogeneous firing rates. In addition, due to the binary character of the underlying bursting state and the high dimensionality of the problem, traditional optimization methods would not be applicable. We solved this problem by applying modern sequential Monte Carlo algorithms (PGAS, Lindsten 2014, for joint estimation for time-varying signals and model parameters) for the first time in the context of spike inference. In summary, the novelty of our work is both in modeling the firing statistics and the inference strategy used.

Another SMC algorithm (Greenberg et al., 2018) stated that the fitted parameters showed some degeneracy, resulting in ambiguous fitting parameters. It would be good to know if this problem was avoided by the authors.

As the reviewer pointed out, one of the weaknesses of the SBM approach is the optimization of the model parameters. This is expected, as SBM uses a biophysical model of the calcium indicator, and a general issue of dynamical models is the presence of so-called sloppy directions in the parameter space, which leads to ambiguous estimations. This is an intrinsic problem due to the model complexity also associated with poorly known parameters such as kinetic constants, which are hard to constrain experimentally. PGBAR uses a much simpler model to describe calcium transients (a second-order autoregressive process) precisely to avoid the non-identifiability of model parameters. Furthermore, we employed a parameterization of the autoregressive model (discussed in the Reparameterization section of Materials and Methods) regarding peak response to a single action potential, decay constant, and rise time (i.e., time to peak). These phenomenological parameters are well documented for different calcium indicators, which enables us to design appropriate prior distributions that significantly facilitate the identifiability of parameters.

Reviewer #2 (Public Review):

Summary:

Methods to infer action potentials from fluorescence-based measurements of intracellular calcium dynamics are important for optical measurements of activity across

large populations of neurons. The variety of existing methods can be separated into two broad classes: a) model-independent approaches that are trained on ground truth datasets (e.g., deep networks), and b) approaches based on a model of the processes that link action potentials to calcium signals. Models usually contain parameters describing biophysical variables, such as rate constants of the calcium dynamics and features of the calcium indicator. The method presented here, PGBAR, is model-based and uses a Bayesian approach. A novelty of PGBAR is that static parameters and state variables are jointly estimated using particle Gibbs sampling, a sequential Monte Carlo technique that can efficiently sample the latent embedding space.

Strengths:

A main strength of PGBAR is that it provides probability distributions rather than point estimates of spike times. This is different from most other methods and may be an important feature in cases when estimates of uncertainty are desired. Another important feature of PGBAR is that it estimates not only the state variable representing spiking activity but also other variables such as baseline fluctuations and stationary model variables, in a joint process. PGBAR can therefore provide more information than various other methods. The information in the GitHub repository is well-organised.

Weaknesses:

On the other hand, the accuracy of spike train reconstructions is not higher than that of other model-based approaches, and clearly lower than the accuracy of a model-independent approach based on a deep network. The authors demonstrate convincingly that PGBAR can resolve inter-spike intervals in the range of 5 ms using fluorescence data obtained with a very fast genetically encoded calcium indicator at very high sampling rates (line scans at ≥ 1 kHz). It would be interesting to more systematically compare the performance of PGBAR to other methods in this regime of high temporal resolution, which has not been explored much.

We appreciate the Reviewer's comment. In response to this observation, we have now included a thorough comparison of PGBAR, MLspike, and CASCADE in addition to the analysis of our cerebellar dataset acquired with a high sampling rate (Figure 9 in the revised preprint). PGBAR and CASCADE predictions are comparable in terms of correlation with the ground truth spikes, and both outperform MLspike. We have also quantified the spike time accuracy as the average distance between ground-truth spikes and the nearest prediction for all the methods. Among the three, PGBAR has the lowest variability of spike time accuracy across our experimental trials. We concluded that while PGBAR and CASCADE show comparable correlations with ground truth, our method provides more reliable spike time estimates.

Recommendations for the authors

Reviewing Editor (Recommendations For The Authors):

In the discussion with reviewers, it was also suggested that while the manuscript emphasized the high temporal resolution of the method (5 ms), this was achieved under favorable conditions (very high sampling rate, fast indicator). Results cannot be compared easily to alternative methods based on published data because these conditions are unusual. Do other methods (at least some of which are presumably easier to use) achieve similar temporal resolution when applied to the same dataset? I feel this could be addressed easily and add valuable information.

We thank the Reviewing Editor for the suggestion. In our revised preprint, we have now added a full comparison between the performance of PGBAR, MLspike (as an alternative

Bayesian approach), and CASCADE (as a state-of-the-art supervised method) tested on our cerebellar dataset. This analysis highlights the improved reliability of our method in terms of temporal accuracy and trial-to-trial variability.

Reviewer #1 (Recommendations For The Authors):

- It is in several places difficult to understand the bigger context of some details. For example, the authors write "In this work, we use Monte Carlo methods to approximate the posterior distribution in Eq. (13)." It would be helpful to state what the bigger goal behind this procedure is, here and at other places. Please go through the Introduction and the Results, there is some room for improvement in terms of accessibility.

We thank the Reviewer for the comment. Monte Carlo methods are generally used when dealing with intractable (non-analytical) probability distributions, which is the case for the models used for spike inference. The “bigger goal behind this procedure” is just the numerical approximation of posterior probabilities, which simply formalizes the question of estimating unknowns from data given a statistical model according to the Bayesian theorem. The advantage of Monte Carlo methods, compared to other techniques (e.g., variational methods), is to be statistically unbiased, which is one of the main reasons why we developed this approach. We clarified the goal of the Monte Carlo inference In the introduction, by adding the following text:

(line 79) “In this work we employ the particle Gibbs (PG) sampler on a bursting autoregressive (BAR) model of time series calcium-dependent fluorescence to provide not only point estimates of spike times but also quantify the statistical uncertainty of each estimate. This is important for downstream analyses such as comparing activity across neurons or conditions.”

We introduce the Results/Model section with the sentence:

(line 91) “To infer spike times and their uncertainty from noisy fluorescence traces, we first build a probabilistic generative model that captures the main dynamics underlying the fluorescence signal.”

And later on in the Results/Sequential Monte Carlo section, we added:

(line 156) “The model described in the previous section is analytically intractable, therefore we employ Monte Carlo methods to sample from the posterior distribution in Eq. (13) of spike times and model parameters, allowing us to make probabilistic inferences rather than relying on point estimates alone.”

In the Abstract: "it provides a flexible statistical framework to test more specific models of calcium indicators". What is meant by this sentence? I was unable to find any results related to this statement.

In our work, we propose a statistical model (depicted in Figure 1A) that accounts for a binary model for non-homogeneous firing, a Gaussian random walk to describe the modulation of the baseline fluorescence coupled to an autoregressive process to link spikes to fluorescence. The phrase quoted by the Reviewer refers to the possibility of replacing the autoregressive model with more specific models of calcium indicators in the future. For instance, employing the biophysical models of calcium indicators to refine the link between spikes and calcium fluorescence. The inference algorithm does not depend on the specific spike-to-fluorescence model. In this sense, our framework is flexible as it offers the opportunity to analyze data acquired using other calcium indicators.

The authors write "One of the key advantages of our sampling algorithm is the joint estimation of latent states and time-independent model parameters." Why is this an advantage? Advantage compared to which alternative algorithm?

We thank the reviewer for this comment. All existing spike inference algorithms use ad-hoc techniques to choose or calibrate the hyperparameters introduced. The estimation of spike times is in general highly sensitive to parameters such as the peak fluorescence in response to single action potentials, kinetic constants, noise levels, baseline, or any regularization or model parameter. These parameters are usually unknown, and all available inference methods provide additional prescriptions to calibrate them. This problem can lead to the propagation of errors and systematic biases. Modern Monte Carlo algorithms, such as the ones employed in our work, address specifically this problem by targeting the joint posterior distribution of all time-dependent variables and the model parameters. Compared to previous approaches, our method offers a statistically rigorous algorithm to identify the parameters. Furthermore, this approach enables us to use Bayesian priors to constrain their ranges without introducing ad-hoc biases and reducing the sensitivity to inaccurate choices of hyperparameters compared to other methods (MLspike), as shown in our new Figure 10 (following a suggestion from Reviewer 2), where we illustrate a parameter sensitivity analysis across MLspike and PGBAR (see responses to Reviewer 2 for further details). We clarified this in the Introduction by adding the sentence:

(line 60) “[...] Moreover, current Bayesian methods do not treat time-independent model parameters (e.g. rate constants) and dynamic variables equally. Instead, they require additional optimization procedures to calibrate model parameters, typically relying on ad-hoc tuning or grid search. This separation can lead to biased inference and poorly calibrated uncertainty estimates, particularly when parameters such as calcium decay time or spike amplitude are inaccurately specified. In contrast, our approach jointly infers both spike times and model parameters within a unified Bayesian framework, enabling uncertainty-aware estimation and avoiding separate, error-prone calibration steps.”

and In the section “Validation and performance of PGBAR” we added the text:

(line 201) “One of the key advantages of our sampling algorithm is the joint estimation of latent states and time-independent model parameters, such as spike amplitude, decay time, noise level, and baseline variance. This stands in contrast to most existing spike inference algorithms, which rely on fixed or externally calibrated parameters. Such fixed-parameter methods are vulnerable to systematic errors when parameter values are uncertain or misestimated. By jointly sampling from the posterior of all variables and parameters, our method propagates uncertainty correctly and mitigates bias due to manual tuning or poor initialization.”

We also added the following text in the discussion:

(line 411) “The estimation of time-independent model parameters is a well-known issue in spike detection algorithms, typically requiring ad-hoc calibration procedures, grid search, or manual settings. Because spike inference is sensitive to parameters such as the calcium response amplitude, rise and decay kinetics, and noise level, errors in these parameters can substantially affect the accuracy of spike time estimates. By jointly sampling model parameters and latent variables, PGBAR eliminates the need for separate calibration and ensures that uncertainty in parameters is propagated to spike time estimates in a principled way. As illustrated in Figure 10, this leads to a more robust inference compared to existing methods like MLspike, which show greater sensitivity to parameter variation. In addition, PGBAR enables the users to calibrate the inference of action potentials by setting prior mean and variance of phenomenological parameters (e.g. rise and decay constants, firing rates, bursting frequencies).”

The authors write "We tested our approach on the fast calcium indicator GCaMP8f (...)". Be more precise. Why exactly were these experiments done, what aspects of the algorithm were supposed to be tested? It is left to the reader to make sense out of these experiments. Please provide the logic of this experiment.

We thank the reviewer for the comment. We developed our method specifically for regimes of high firing rates. For this reason, in addition to the CASCADE benchmark dataset, we have tested our approach on recordings of cerebellar granule cells due to their fast spiking patterns. For this purpose, we have employed the ultrafast state-of-the-art calcium indicator GCaMP8f combined with linescan imaging techniques to enable fast acquisition rates. We added the following text in the manuscript to clarify:

(line 306) "We tested our approach on the fast calcium indicator GCaMP8f by performing high-speed (2.8 kHz) two-photon linescan calcium imaging of cerebellar granule cells in vitro. GCaMP8f was expressed in the Crus I region of the cerebellum using adeno-associated virus (AAV) injection (Fig. 7A). Compared to GCaMP6f, GCaMP8f exhibits a rise time that is nearly an order of magnitude faster, which we expected to translate into substantially improved temporal accuracy in spike time detection."

The authors write "If we consider as reference correlation the average across the CASCADE dataset (0.75) (...)". Why would this threshold be appropriate? This sounds arbitrary; this experiment was conducted with 2.8 kHz line scan imaging of GCaMP8, while the reference stems from low-rate imaging of older indicators.

We thank the reviewer for the remark. In the sentence quoted, we have used 0.75 as a reference for the state-of-the-art correlation between ground truth and predicted spikes and indicated the lowest temporal resolution (10 ms) where the PGBAR correlation is larger than the reference value. As the Reviewer correctly pointed out, the reference 0.75 refers to datasets with much lower acquisition frequency; therefore, in our revised preprint, we have added a comparison of the correlations obtained from PGBAR, CASCADE, and MLspike using high-speed recordings of cerebellar GCs (Figure 9), showing the increased performance of our method at high temporal resolution.

How was PGBAR evaluated using a given dataset in Figure 4c or in Figure 7? It is unclear to the reviewer whether the priors were automatically/manually adjusted for each data set.

We thank the Reviewer for this comment. Briefly, for the CASCADE dataset, we have designed the priors for all parameters according to the existing characterization of the calcium indicator used in each experiment (Chen et al. 2013). For our cerebellar data, we have performed single stimulation trials for each recording, which we used to design priors on peak fluorescence response, decay constant, and time to peak fluorescence. In the Results section of the revised preprint, we clarified more specifically how priors were designed for the CASCADE and our cerebellar datasets. We have added the following statements:

(line 239) "Bayesian priors for all PGBAR parameters were adapted to each experiment according to the existing characterization of the different calcium indicators used (Chen et al., 2013)."

(line 314) "For each recorded soma and bouton we applied two types of stimulations. Single time point stimulation and a fixed stimulation pattern generated from a 20 Hz Poisson process with 29 stimulation time points. First, we used the single-stimulation trials to design prior distributions of amplitudes, rise and decay constants (Fig. 7C). Next, we used PGBAR to analyze independently each Poisson stimulation trial in Figure 7E. By generating thousands

of posterior samples of spike time patterns, we obtained the spike probability for all time frames and trials (Fig. 7F)."

The authors write "This analysis illustrates the variability expected when analyzing multiple trials of the same neuron." Variability across trials of neuronal activity? Or variability of spike inference?

We thank the reviewer for the comment. In the revised text, we clarify that we refer to the variability of spike inference across trials.

The original statement has been revised to read:

(line 301) "This analysis illustrates the expected variability of spike inference when analyzing multiple trials of the same neuron."

Technical question: How can the authors be sure that glass electrode stimulation only elicits a single AP per stimulation? This was not clear to me from the manuscript alone.

We thank the reviewer for the question. Our experimental protocol is designed in a way that in each trial we make sure a single electrical stimulation elicits a single AP. We adjust our stimulation strength until we see an all-or-none calcium transient in response to a single AP. Given the fast temporal properties of GCaMP8f, we could distinguish a single AP response from multiple APs during a single electrical stimulation. We then introduced a single stim trial ahead of each Poisson-train trial to see whether our stimulation strength could elicit a single AP response reliably and consistently. In this way, we ensured that every single stim was producing a single AP.

Figure 8: Please explain what you mean by "bouton". What is the dashed line in (A)? Why is it interesting to look at the differences between bouton and soma?

We thank the Reviewer for the comment. In the updated text we clarified that we refer to synaptic boutons along the parallel fiber (line 311) and that the dashed line in Figure 8 refers to the ground-truth number of spikes (29). We also pointed out that the estimated delay between somas and boutons is compatible with the proximity of synaptic boutons to the stimulation site along the parallel fiber by adding the following text:

(line 340) "This result is compatible with the proximity of synaptic boutons to the electrical stimulation along the parallel fiber. We analyzed both signals from somata and synaptic boutons because in vivo two-photon imaging can be made from both parts of the cell. Here we showed that our method performs reliably on both, demonstrating its robustness across recording sites."

Reviewer #2 (Recommendations For The Authors):

The authors emphasised the result that PGBAR can resolve spike timing differences of 5 ms. However, this result was obtained based on fluorescence data measured with a very fast calcium indicator at very high sampling rates. It remains unclear how the performance of PGBAR compares to other methods in this regime of high temporal resolution, which has not been explored much in previous comparisons of methods. Potential users interested in this regime would benefit from a direct comparison to other approaches.

We thank the Reviewer for this suggestion. In our revised manuscript, we have included a detailed comparison of spike time accuracy between our method, MLspike, and Cascade, summarized in Figure 9. In this analysis, we used our in vitro dataset to estimate the average

temporal accuracy of spike detection across the three methods. As discussed in the main text, the average temporal accuracy was defined as the temporal offset between the ground truth and the nearest detected spikes averaged across the ground truth. The distributions of temporal accuracies across our experiments obtained from MLspike, Cascade, and PGBAR differ in their spread, with 10th-to-90th percentile ranges of 14 ms, 8 ms, and 3 ms, respectively. This result demonstrates that PGBAR estimates are more reliable than MLspike and CASCADE across trials, with a narrower unbiased distribution of temporal accuracy.

In practice, approaches are more appealing to users when they do not require dedicated measurements to estimate parameters such as rise/decay time constants of calcium fluorescence signals within cells. Users may therefore be interested to know how results would be affected if these parameters are estimated only crudely. It would thus be useful to know how spike probability estimates vary as a function of these parameters, which should be easy to test systematically, and whether the sensitivity of PGBAR to inaccurate initial parameter estimates is lower or higher than that of other methods, which should also be easy to test. As PGBAR jointly estimates spike probabilities and model parameters, it may have an advantage here over other methods.

We thank the Reviewer for this suggestion. In the new Figure 10, we show a parametric sensitivity analysis for both PGBAR and MLspike. For PGBAR, we considered the hyperparameters of the Bayesian priors associated with the peak response to a single spike and the baseline variance, which influences how much of the fluorescence can be attributed to baseline modulation. For MLspike, we considered the transient amplitude and the decay time constant. For both methods, we varied the parameters between -50% and +50% of their optimal value and estimated the correlation between predictions and ground truth as well as the number of spikes (Figure 10A). Next, we calculated the coefficient of variation across all parameter configurations for each trial (Figure 10B). Our analysis shows that, compared to previous methods, PGBAR has a much lower sensitivity to the initial choices of the hyperparameters, confirming the intuition of the Reviewer thanks to the simultaneous inference of spike times and model parameters. This result provides an important addition to our work.

Equation 10: -1 should be in subscript (t-1). Remark: I have not fully verified the mathematical parts because some of it is beyond my expertise.

We thank the Reviewer for pointing out the typo. This has been corrected in the revised preprint.

$$P(X_{1:T}, F_{1:T} | \theta) = \mu^\theta(X_1) \cdot \prod_{t=2}^T f_t^\theta(X_t | X_{t-1}) g_t^\theta(F_t | X_t) \quad (10)$$

<https://doi.org/10.7554/eLife.94723.2.sa0>